

Wireless Networks

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Wireless Networks

Lecture-1

Introduction to Wireless Networks



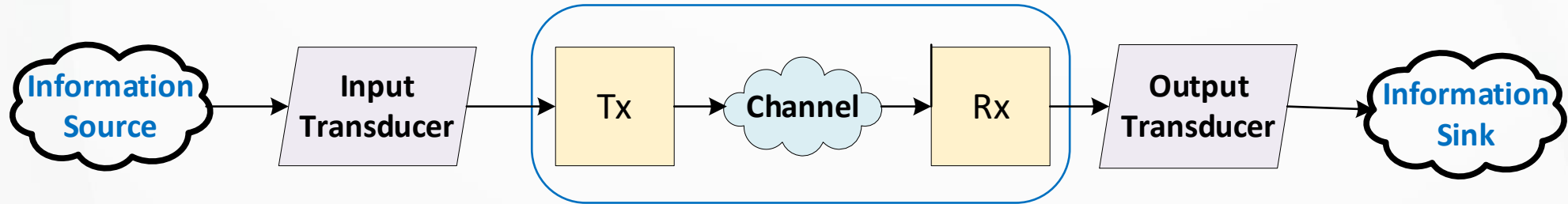
Content

- ☑ Digital Communication System Elements
- ☑ Layered Network Architecture
- ☑ Wireless vs. Wired
- ☑ Elements of a wireless network
- ☑ Frequency, Wavelength, Amplitude, and Phase
- ☑ Time and Frequency Domains
- ☑ Electromagnetic Spectrum



Introduction: Digital Communication System Elements

- ❖ The purpose of digital communication system is to **send** information from **source** to **sink**



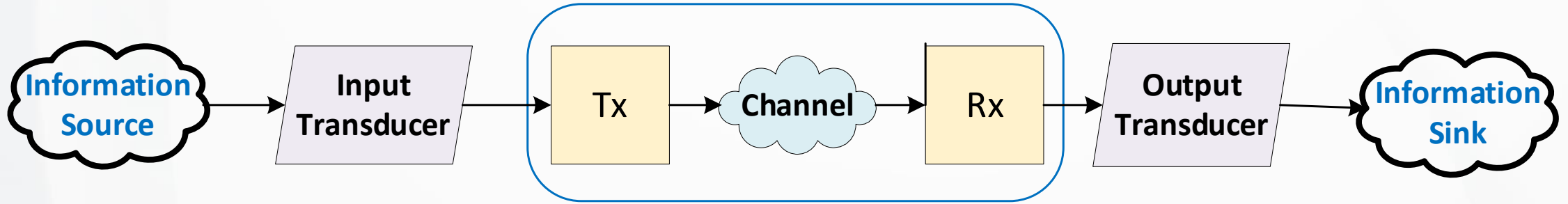
❖ Transmitter (Tx)

- Convert a signal to **another waveform** that are **suitable** to transmission over a particular channel.



- The **signal** coming from the transducer is a **baseband**, **lowpass** signal that is usually **bandlimited**. For instance, voice bandwidth is 4kHz, which cannot be transmitted over the channel due to its bandlimited nature.
- The transmitter (Tx) translates the signal's frequency band to another frequency suitable for transmission over the channel.

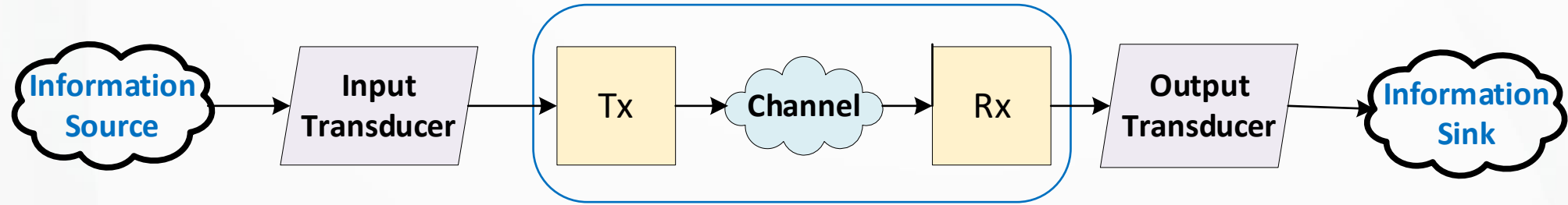
Introduction: Digital Communication System Elements



❖ Transmitter process include:

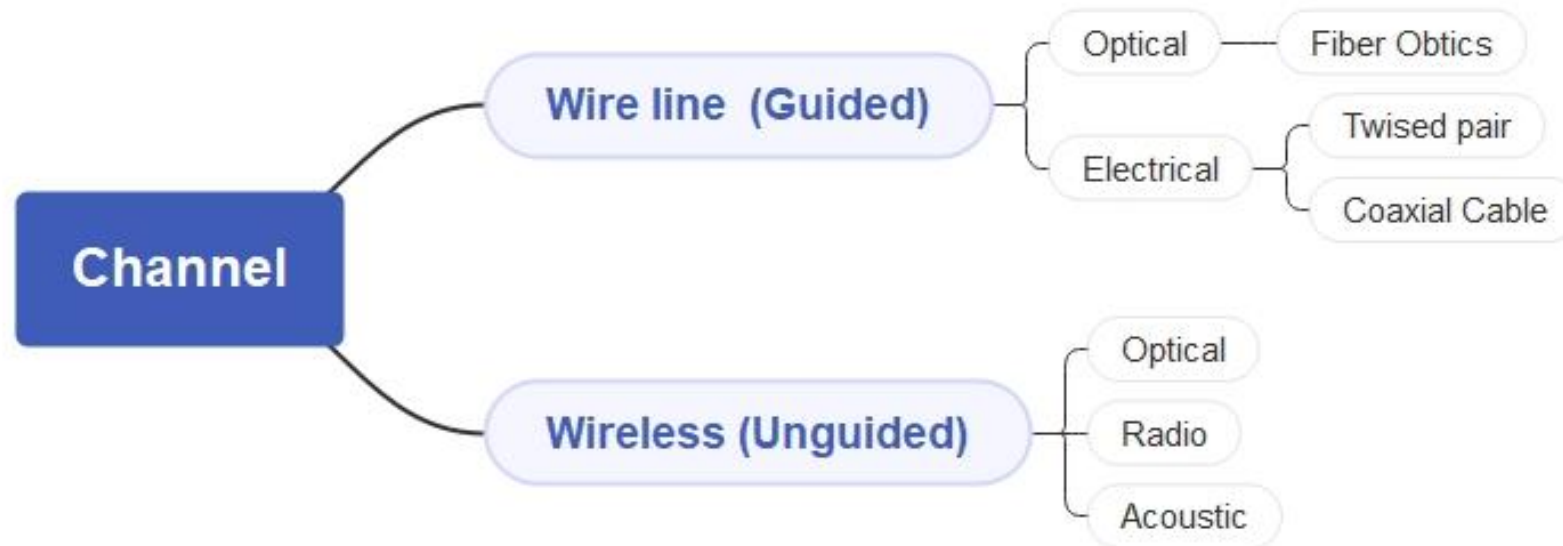
- Signal formatting
- Sampling and quantization
- Source encoding
- Encryption
- Channel encoding
- Multiplexing
- Pluses shaping
- Modulation
- Amplification and filtering
- Frequency translation

Introduction: Digital Communication System Elements



❖ Channel

Channel is a physical medium used to send signal from **Tx** to **Rx**.



Introduction: Digital Communication System Elements

❖ Channel

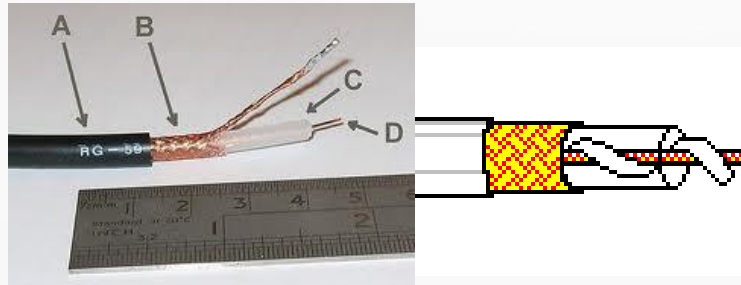
Twisted Pair (TP)

- two insulated copper wires
 - Category 3: traditional phone wires, 10 Mbps Ethernet
 - Category 8: 25Gbps Ethernet
- Shielded (STP)
- Unshielded (UTP)



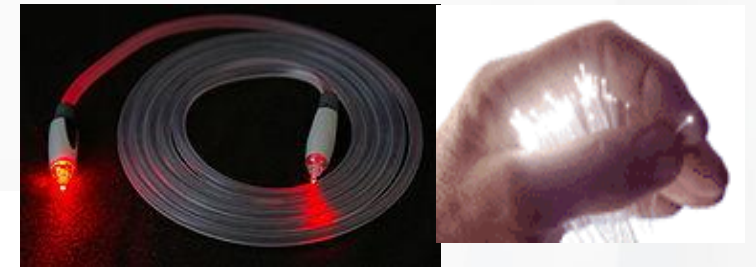
Coaxial cable:

- two concentric copper conductors
- bidirectional
- baseband:
 - single channel on cable
 - legacy Ethernet
- broadband:
 - multiple channels on cable
 - HFC (Hybrid Fiber Coax)



Fiber optic cable:

- high-speed operation
- point-to-point transmission
- (10' s-100' s Gbps)
- low error rate
- immune to electromagnetic noise

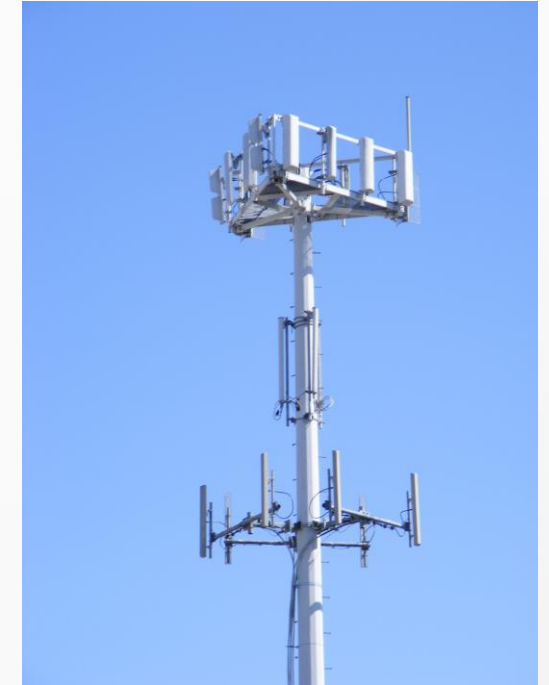


Introduction: Digital Communication System Elements

❖ Channel

Radio link types:

- ❑ terrestrial microwave
 - ❖ e.g. 90 Mbps channels
- ❑ LAN (e.g., Wifi)
 - ❖ 11Mbps, 54 Mbps, 600 Mbps
- ❑ wide-area (e.g., cellular)
 - ❖ 5G cellular: ~ 40 Mbps - 10Gbps
- ❑ satellite
 - ❖ 27-50MHz typical bandwidth
 - ❖ geosynchronous versus low altitude
 - ❖ For geosync - 270 msec end-end delay to orbit

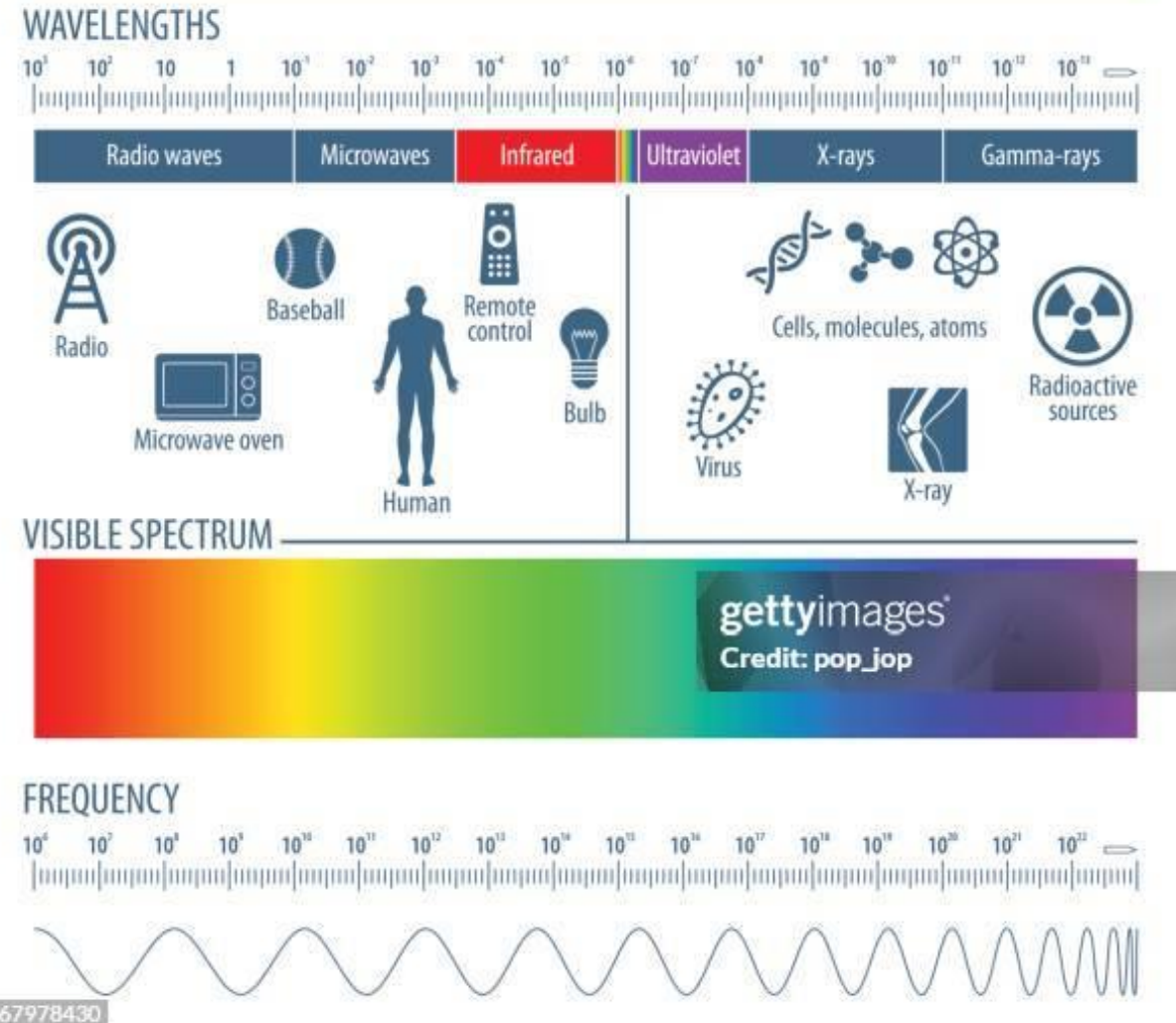


Introduction: Digital Communication System Elements

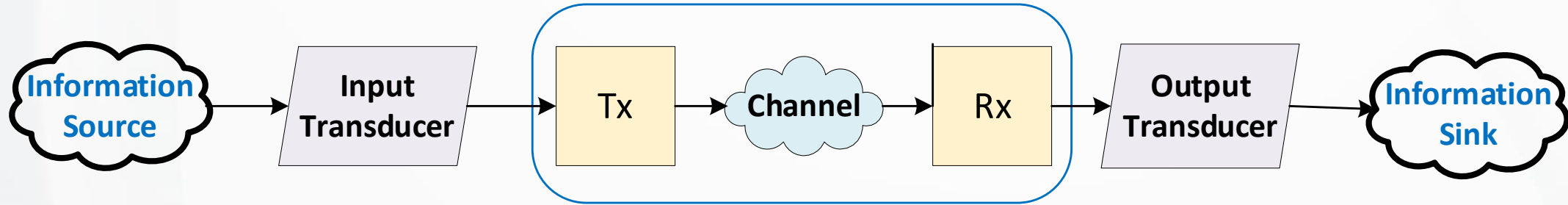
Licensed vs. Unlicensed Bands

- Licensed bands
 - Operators get the license by paying money
 - 800 MHz, 1.8 GHz, 2.1 GHz
- Unlicensed bands
 - Used without license as long as the regulatory requirements are met such as maximum transmit power level, spectral mask, etc.
 - 900 MHz, 2.4 GHz, 5 GHz, 60 GHz ISM (industrial, scientific and medical) bands
- Different for different countries

ELECTROMAGNETIC SPECTRUM



Introduction: Digital Communication System Elements



❖ Channel

The common property of these types is what ever I put on the channel I get something different: **Signal is distorted**

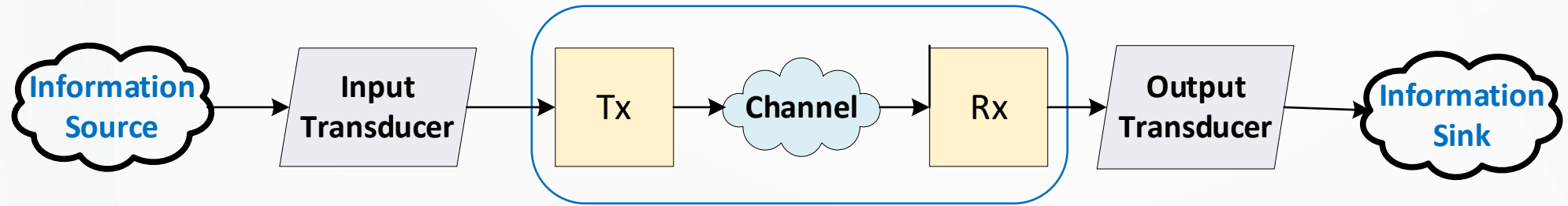


➤ Two types of distortions

1) **Additive**: Output of the channel contains other signals plus input signal.

2) **Non additive**: like (spectral modification, filtering, non-linear distortion, ...)

Introduction: Digital Communication System Elements

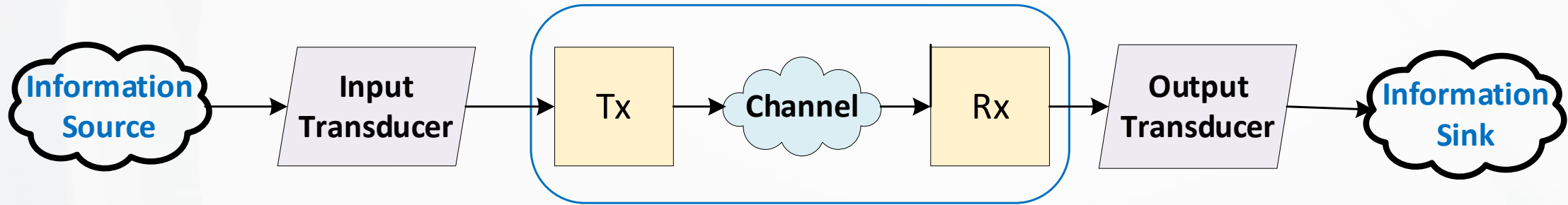


❖ Channel

Why modern communication is digital ?

- **In analog** , the signal is just amplified and this amplifies the noise combined with original signal.
- **In the digital**, the receiver **Rx** try to answer if the received signal is 0 or 1 ? . So, upon **Rx** knows 1, it cancel the entire signal and replace it with the signal 

Introduction: Digital Communication System Elements



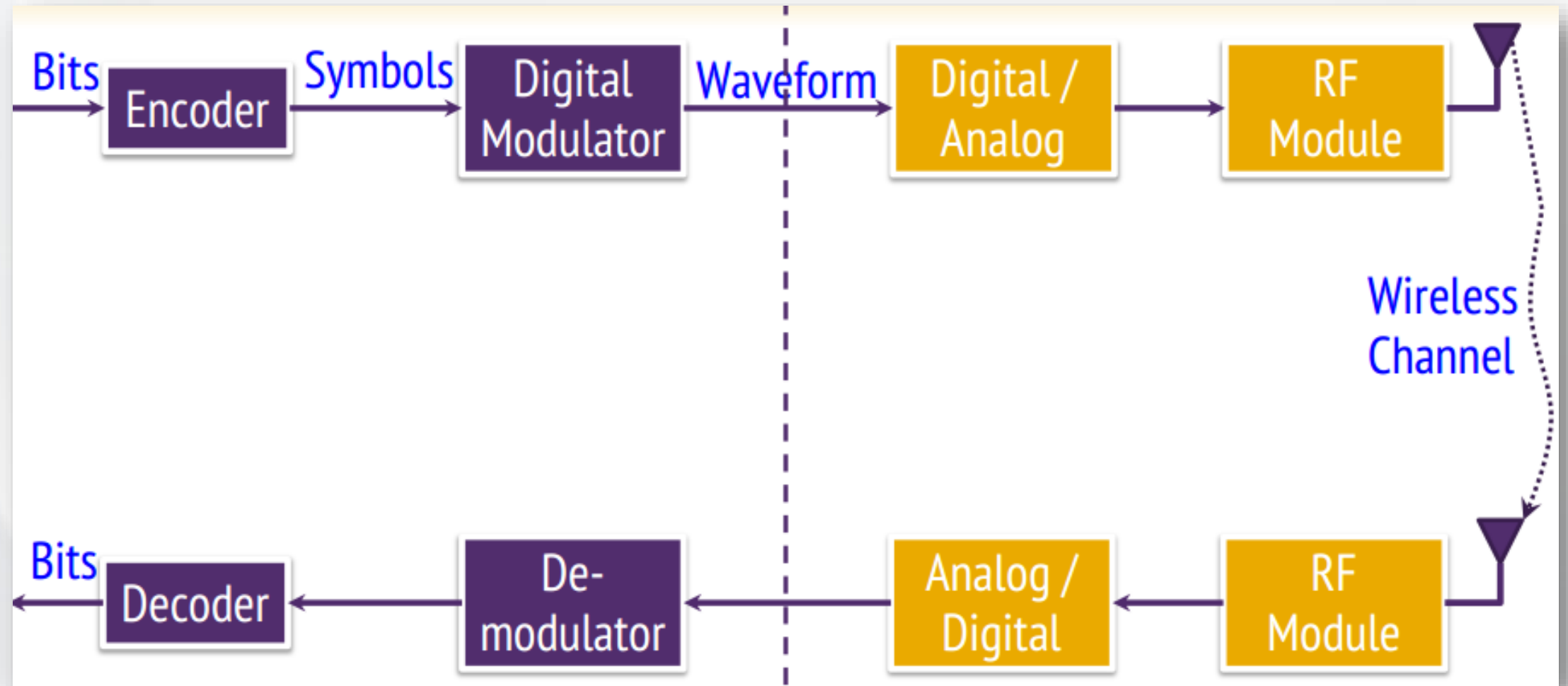
❖ Receiver (Tx)

- Recovers the original signal that passed to the transmitter

❖ Receiver process include:

- Frequency translation
- Amplification and filtering
- Demodulation
- Demultiplexing
- decoding & error correction
- Deciphering
- Reconstruction
- Formatting.

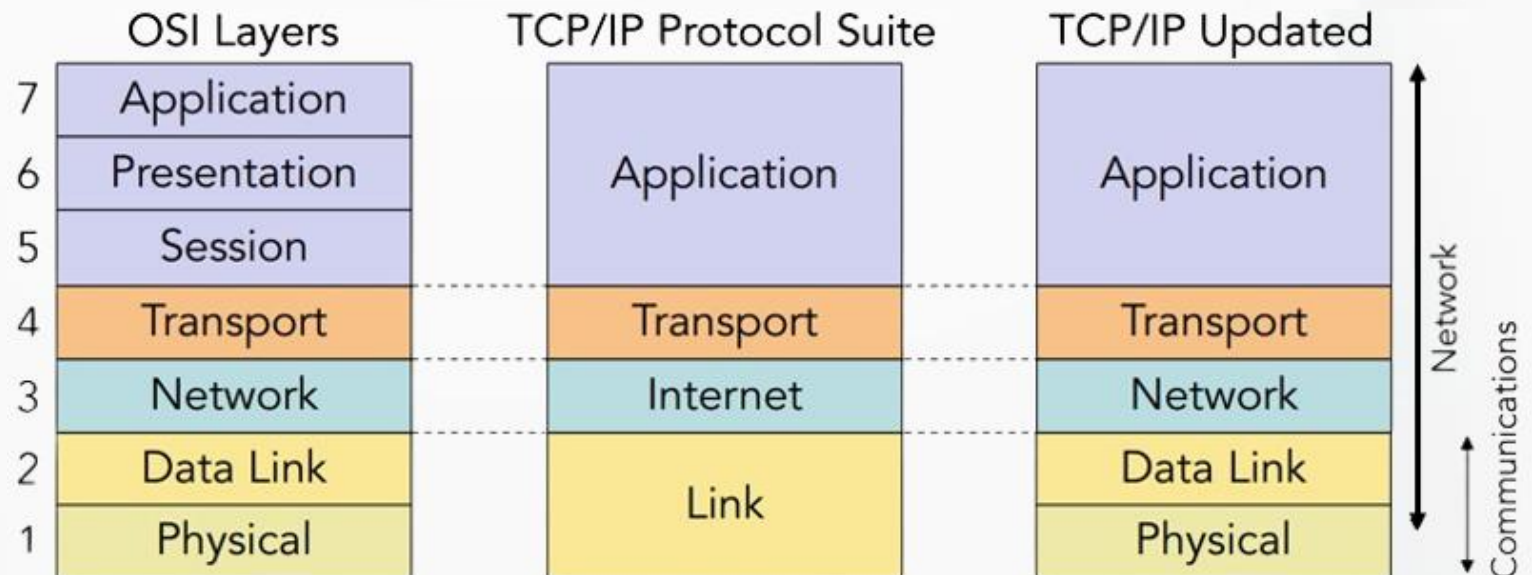
Introduction: Digital Communication System Elements



Introduction: Layered Network Architecture

- **Physical (PHY) Layer:**

1. Deals with the actual transmission of bits over the medium, including modulation, coding, and signal processing.
2. The design is tailored to the specific requirements and constraints of the system. For instance:
 - **WLAN:** Employs techniques like OFDM for short-range high-speed data.
 - **LTE/5G:** Focuses on broader coverage and high-speed data transmission with more complex modulation schemes.

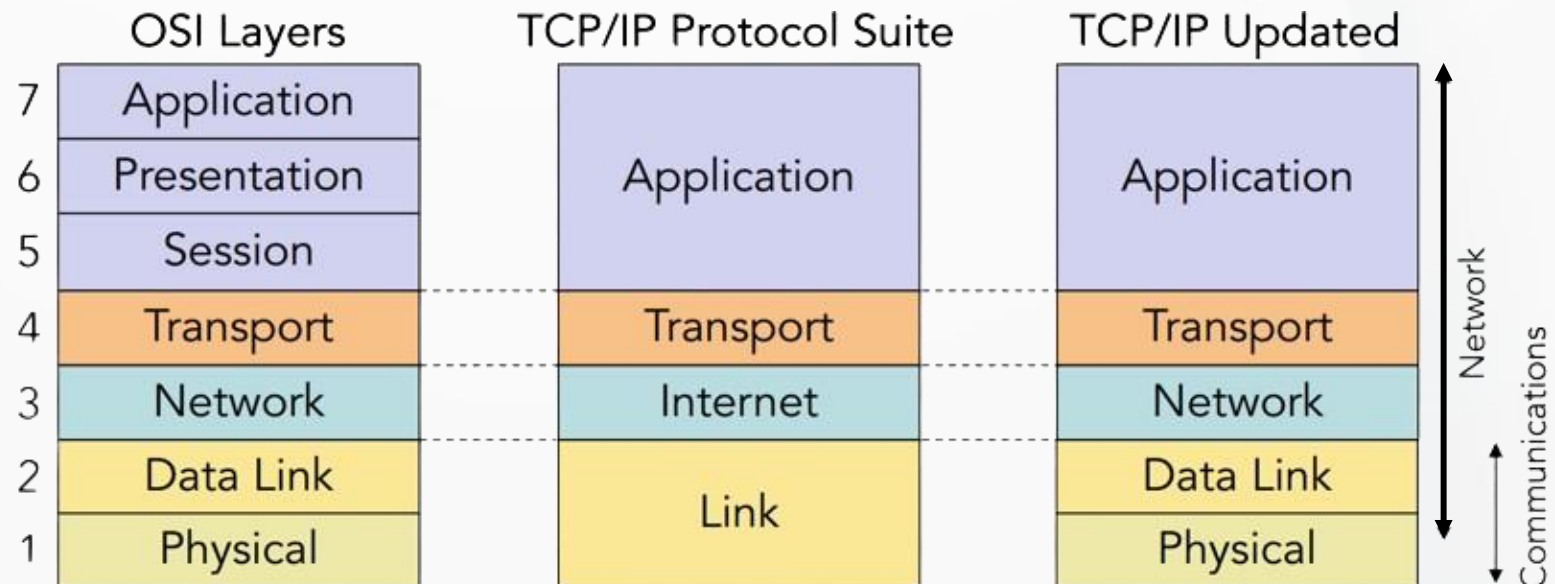


Introduction: Layered Network Architecture

- **Data Link Layer (including MAC):**

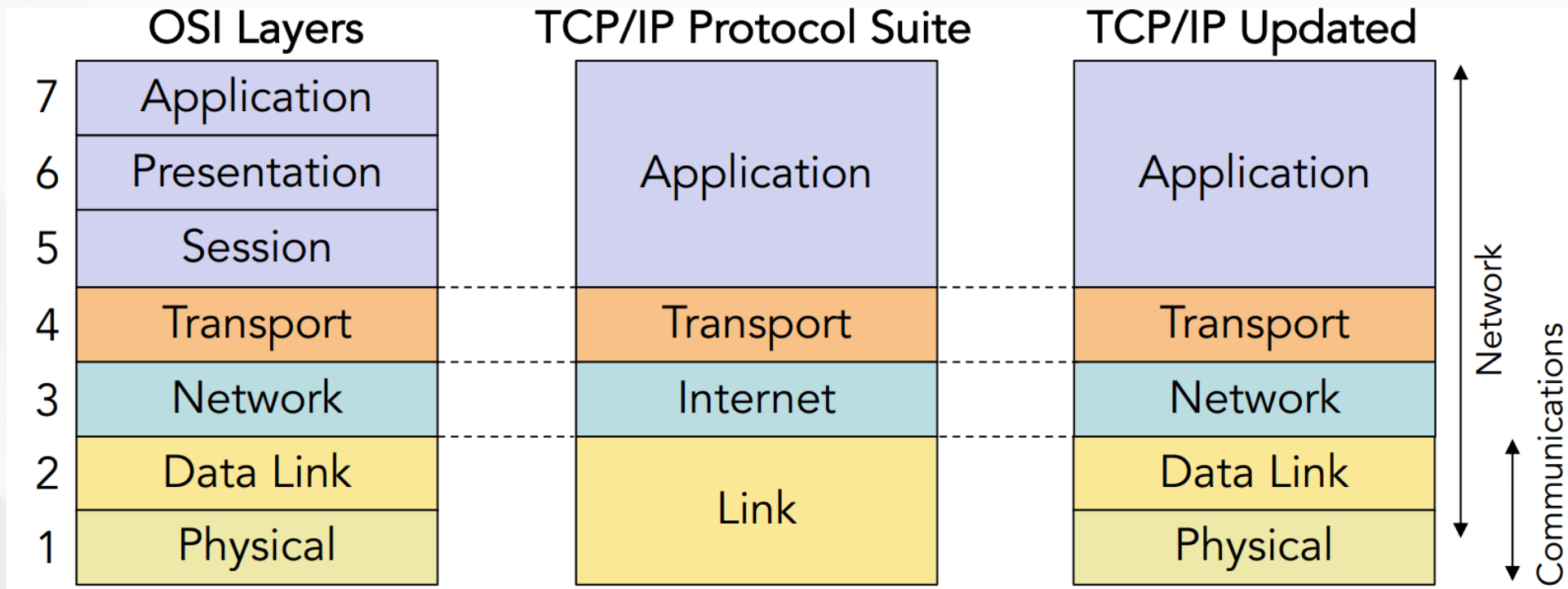
1. Handles data framing, error detection, and Medium Access Control (MAC).
2. MAC (Medium Access Control) manages how devices access and share the communication medium (e.g., airwaves in wireless systems).
3. Its implementation varies significantly based on the type of network. For example:

- ✓ **WLAN:** Uses protocols like CSMA/CA for collision avoidance.
- ✓ **LTE/5G:** Uses advanced scheduling algorithms for efficient spectrum utilization.



Introduction: Layered Network Architecture

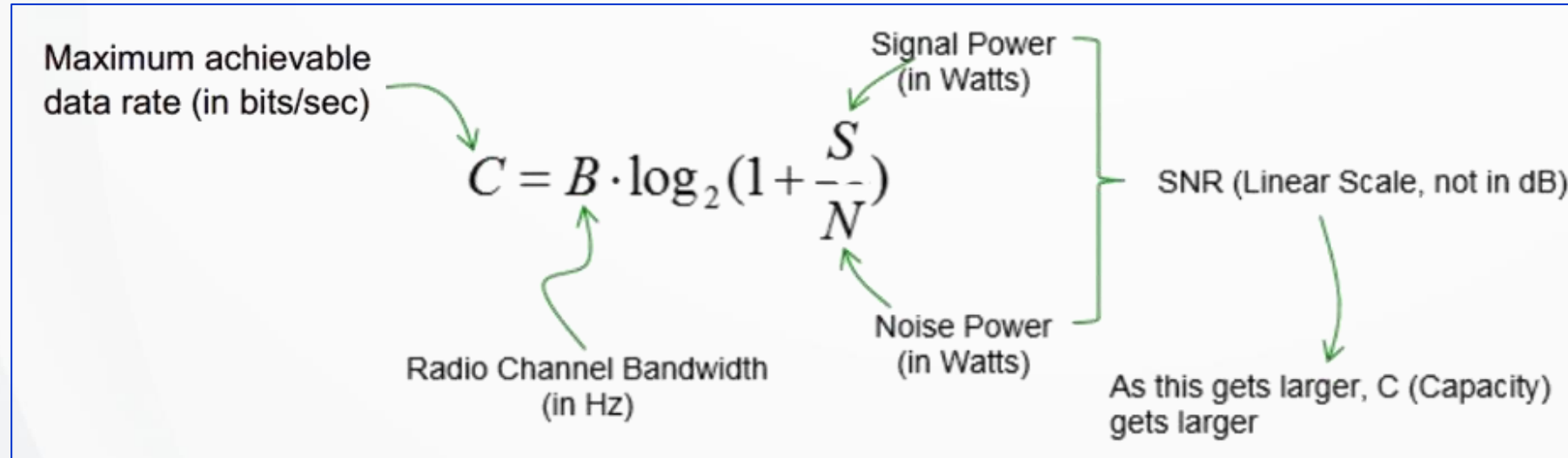
In summary, both layers are customized to optimize performance for specific systems like WLAN, LTE, or 5G, as each has unique requirements for bandwidth, latency, and efficiency.



Introduction: Wireless vs. Wired

❖ Scarce resources, i.e., bandwidth

- ✓ 1 Gbps WLAN vs. 1 Gbps LAN



❖ Less-reliable communication

- ✓ Fading, shadowing, background noise
- ✓ Interferences - inter-symbol interference (ISI), inter-cell interference, interference from other systems

❖ User mobility

- ✓ Handoff, location management
- ✓ Channel reliability affected by Doppler shift (coherence time)

Introduction: Wireless vs. Wired

❖ Channel Gain

❖ **Wireless Channel Gain** measures how the wireless channel alters a transmitted signal's power, accounting for effects like attenuation, amplification, and distortion.

❖ Channel gain may be:

- A **real** number (for amplitude gain).
- A **complex** number (for amplitude and phase gain).

Introduction: Wireless vs. Wired

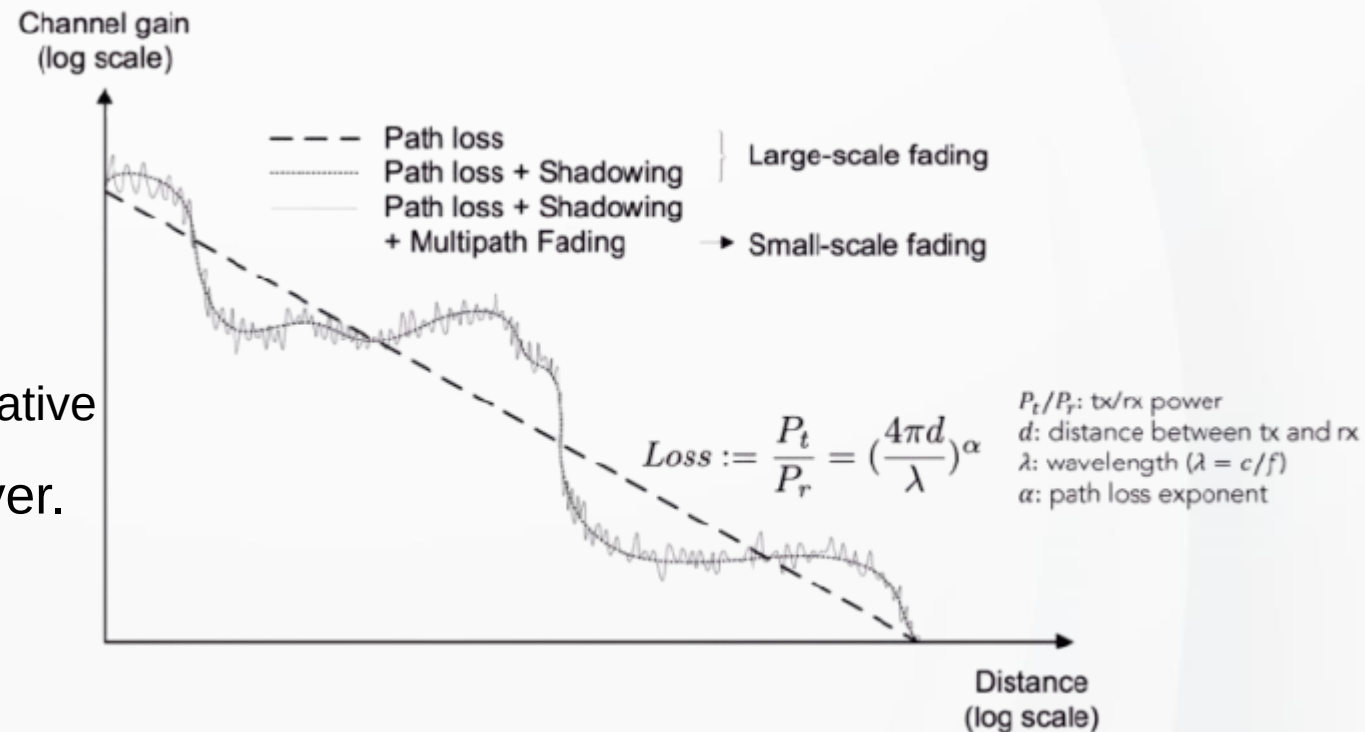
❖ Channel Gain

▪ Factors Influencing Channel Gain:

- ✓ **Path Loss:** Signal attenuation due to the distance between transmitter and receiver.
- ✓ **Shadowing:** Obstructions (e.g., buildings or trees) causing additional signal loss.

✓ **Multipath Fading:** Signal reflections, refractions, and diffractions creating constructive or destructive interference.

✓ **Doppler Shift:** Frequency shifts due to relative motion between transmitter and receiver.



Introduction: Wireless vs. Wired

❖ Channel Gain

❑ Types of Gain:

1. **Large-scale gain:** Accounts for path loss and shadowing; changes slowly over distance.
2. **Small-scale gain:** Accounts for rapid fluctuations due to multipath and motion.

❑ Applications:

1. Channel gain is critical for designing wireless systems, as it directly impacts:
 1. **Signal-to-noise ratio (SNR).**
 2. **Data rate.**
 3. **Error performance.**
2. Used in link adaptation, beamforming, and power control.

Introduction: Wireless vs. Wired

- Time-varying environment
 - Time-varying channel and user mobility
 - Time-varying interferers
 - Location-dependent errors
- Broadcast nature of channel
 - Multiple access for sharing the medium
- Less-secure environment
 - Due mainly to the broadcast nature of the wireless
- Wireless node can be static and fixed
 - e.g., Fixed wireless local loop (WLL) or IEEE 802.16 Broadband wireless access (BWA)
- Mobile node can use a wireline networking
 - e.g., laptop with Ethernet link

Introduction: Wireless & Mobile Networks (WMN)

- Nomadic systems
 - Communications is typically done while the node is stationary
 - WLAN, WPAN
- Mobile systems
 - Communications can be done while the node is moving fast
 - 3G/4G cellular systems
- Nomadic system can provide faster link!

Introduction: WMN: Infrastructure vs. Ad hoc

□ Infrastructure-based wireless networks

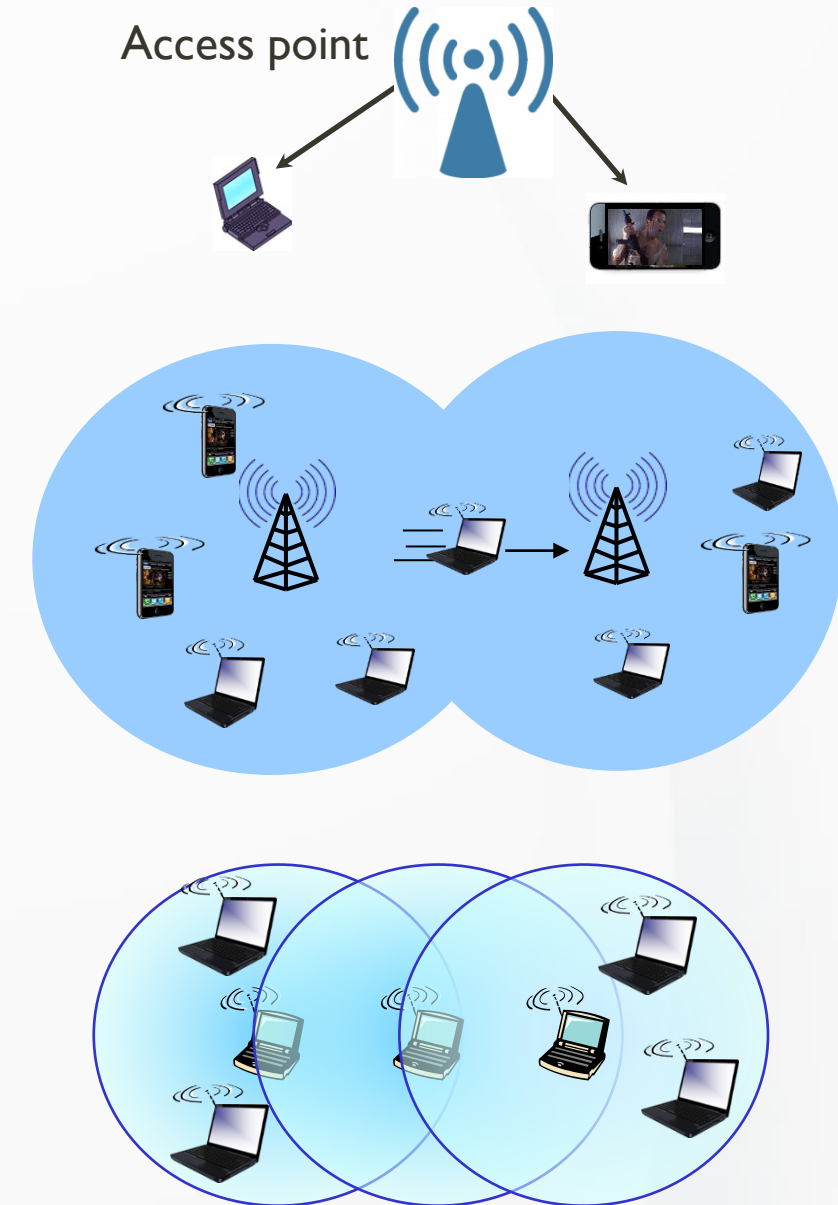
- An access point (AP) or base station (BS) as an interface between wireless and (wireline) backbone
- Star topology (in a cell), handoff support
- Requires cell planning (with frequency reuse)
- Cellular systems, typical 802.11 WLANs with APs, ...

□ Ad hoc networks

- Wireless multi-hop transmission
- Peer-to-peer topology
- 802.11 ad hoc mode, Bluetooth, ...

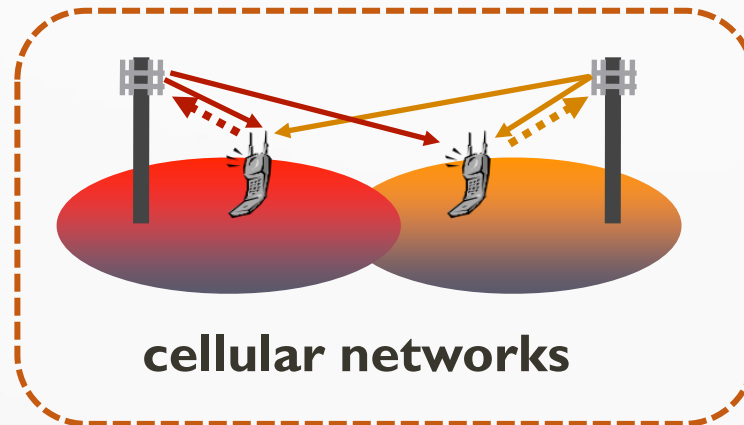
□ Wireless mesh networks

- Network for wireless infrastructure

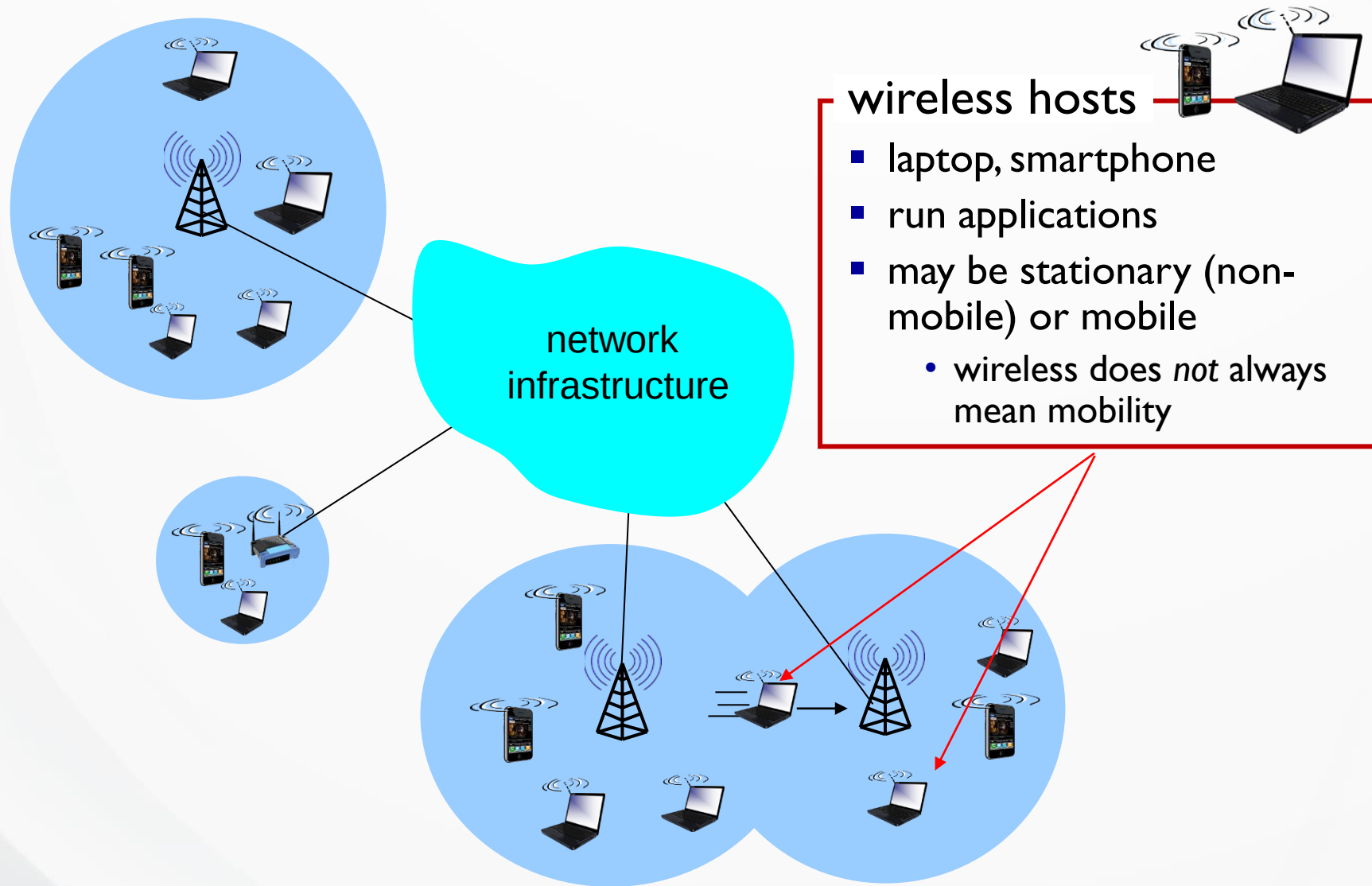


Introduction: Wireless networks

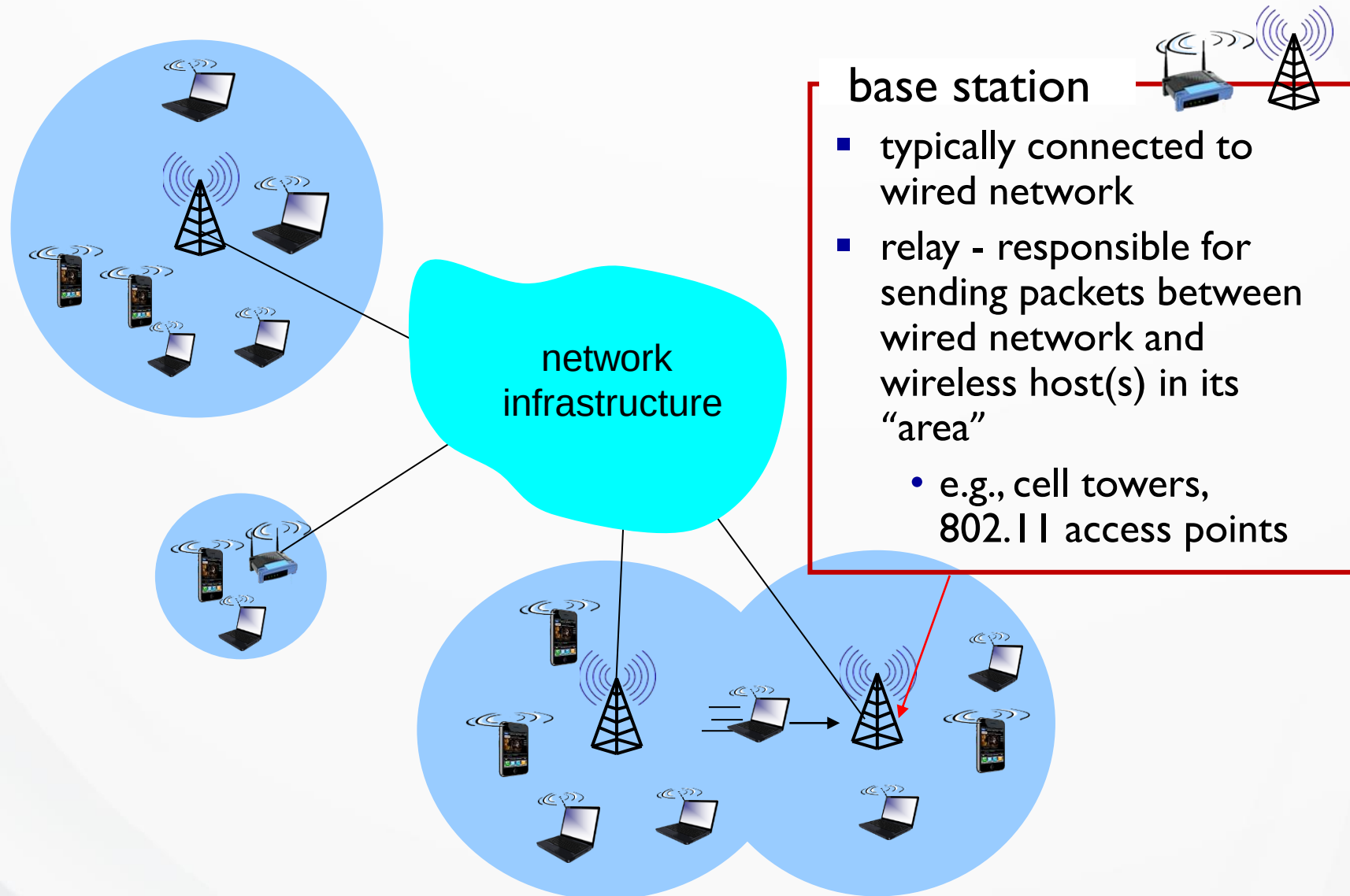
- ❖ Wireless networks are telecommunications networks that allow devices to connect and communicate **without physical connections** like cables.
- ❖ They use radio waves or infrared signals to transmit data between devices.
- ❖ This technology enables users to **move freely within a network area** while maintaining a connection to the network.



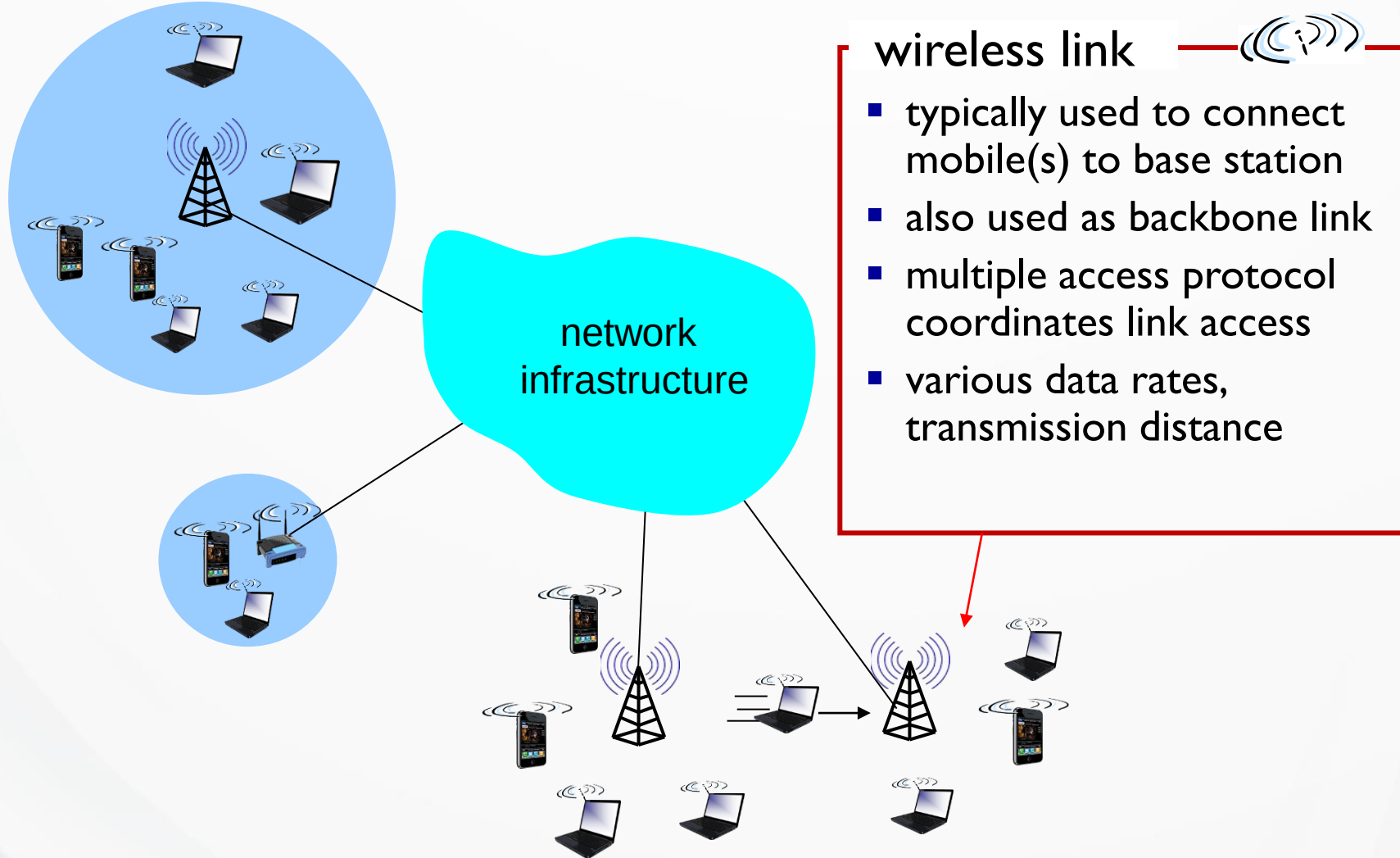
Introduction: Elements of a wireless network



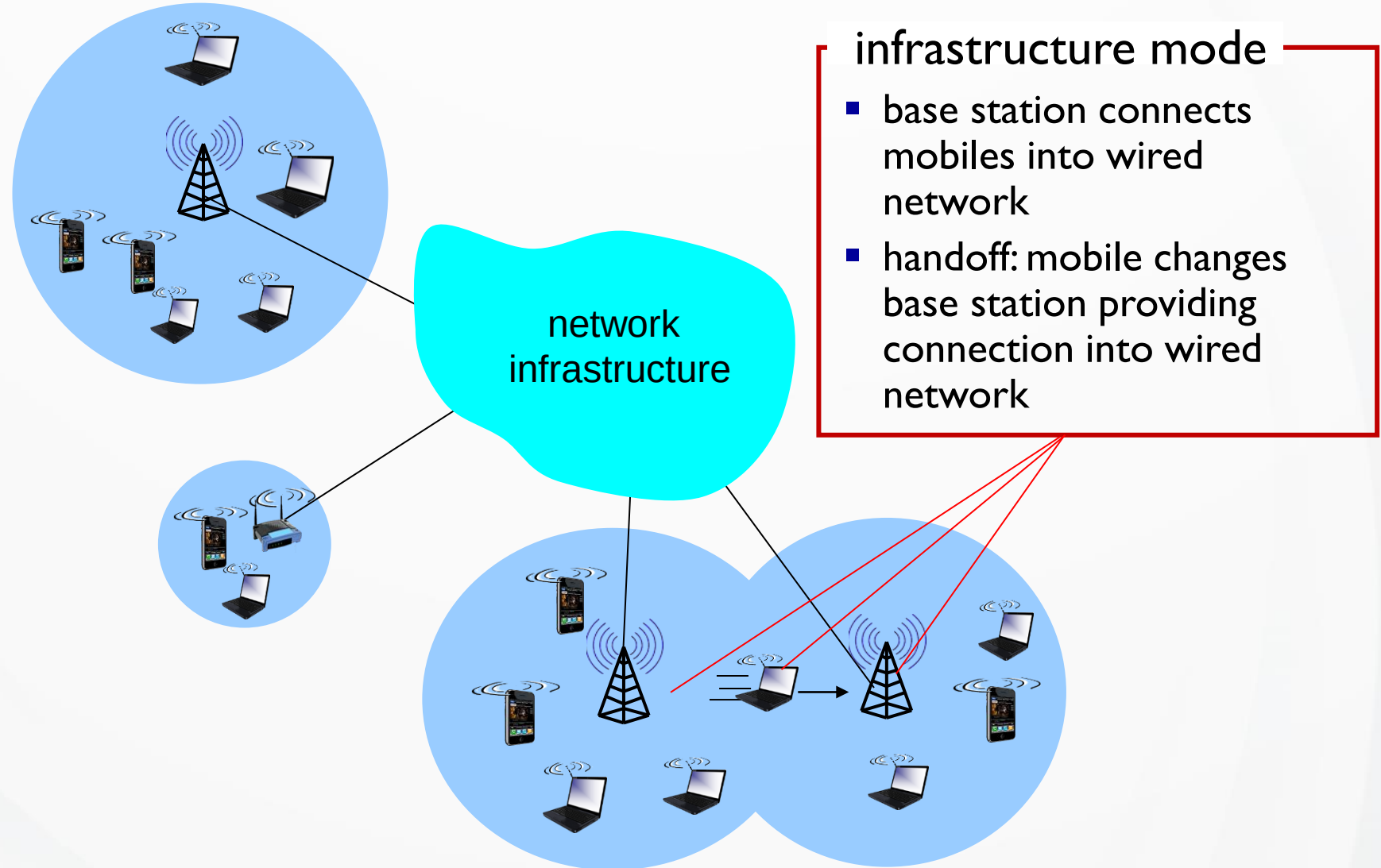
Introduction: Elements of a wireless network



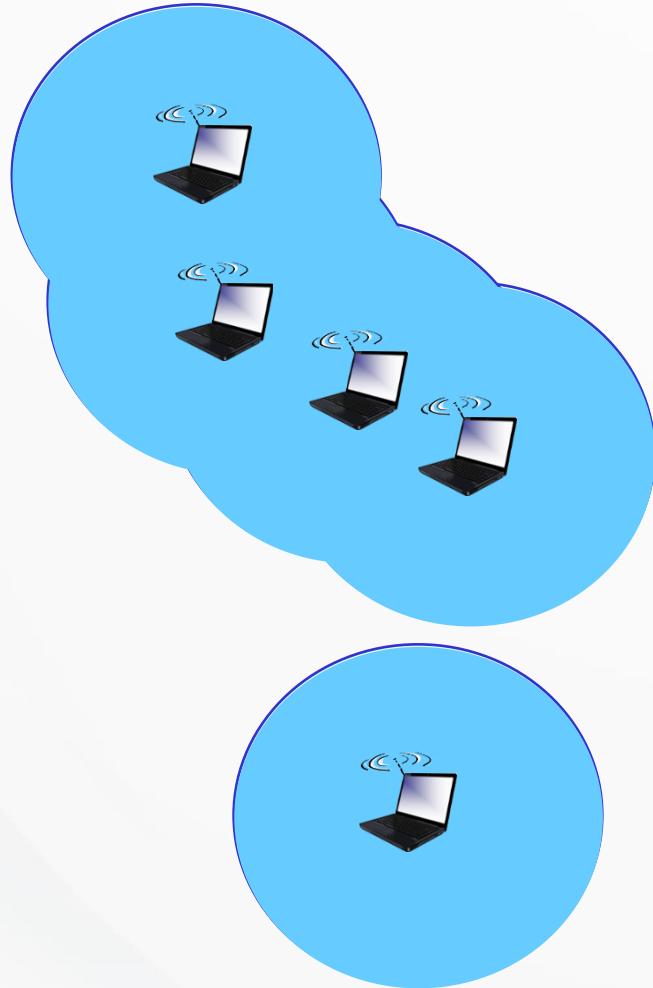
Introduction: Elements of a wireless network



Introduction: Elements of a wireless network



Introduction: Elements of a wireless network



ad hoc mode

- no base stations
- nodes can only transmit to other nodes within link coverage
- nodes organize themselves into a network: route among themselves

Introduction: Wireless network taxonomy

	single hop	multiple hops
infrastructure (e.g., APs)	host connects to base station (WiFi, WiMAX, cellular) which connects to larger Internet	host may have to relay through several wireless nodes to connect to larger Internet: <i>mesh net</i>
no infrastructure	no base station, no connection to larger Internet (Bluetooth, ad hoc nets)	no base station, no connection to larger Internet. May have to relay to reach other a given wireless node MANET, VANET

Frequency, Wavelength, Amplitude, and Phase

- ❖ **Definition:** **Coding** and **modulation** map digital information to signals, enabling retrieval by a receiver using a decoder and demodulator.
- ❖ **Importance:** **Coding** and **modulation** directly impact the capacity and data rate of communication systems.
- ❖ **Innovation:** Continuous development of new coding and modulation techniques to meet growing mobile data demands.

Frequency, Wavelength, Amplitude, and Phase

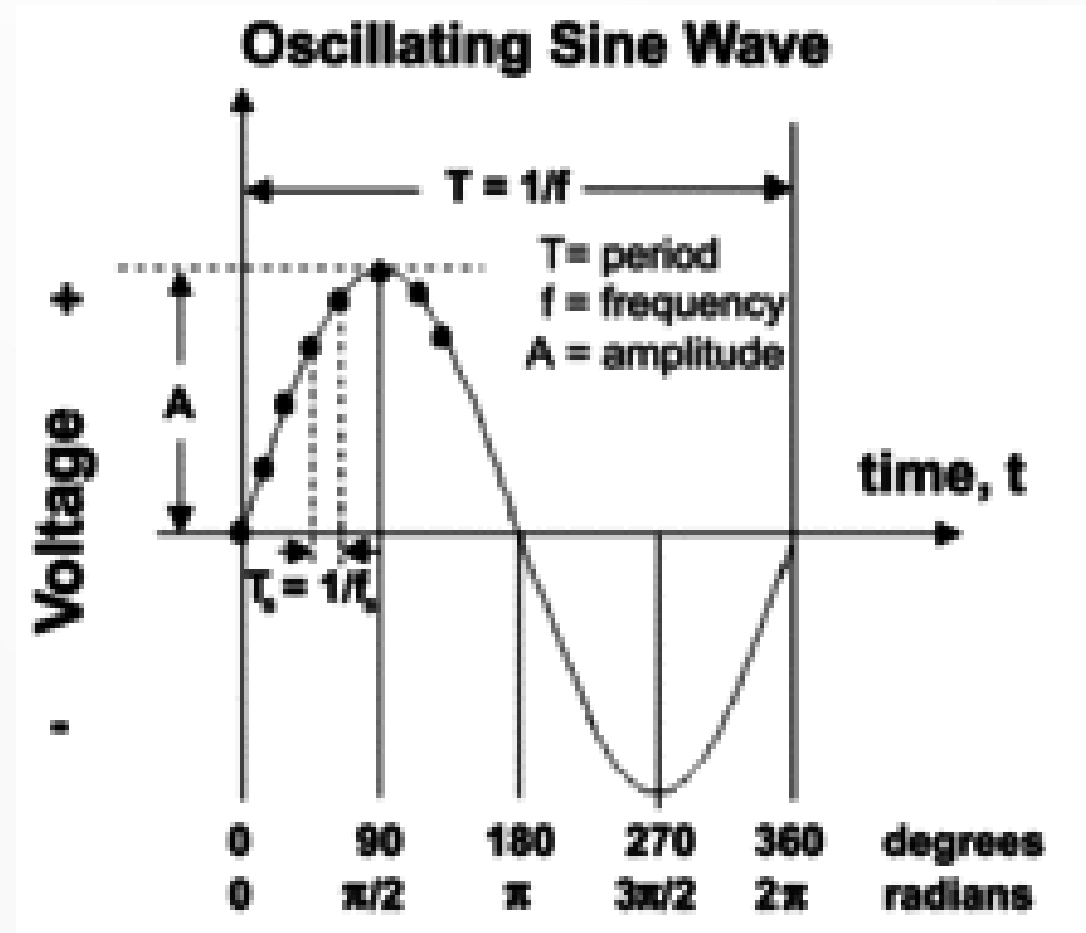
- ❖ Signal waveforms are fundamental carriers of data in communication systems.
- ❖ Understanding waveforms is crucial for communication technology.
- ❖ Waves created by a rock in water are similar to **electromagnetic** waves in wireless networking.
- ❖ **Electromagnetic** waves are generated and received using electronic circuits.



Frequency, Wavelength, Amplitude, and Phase

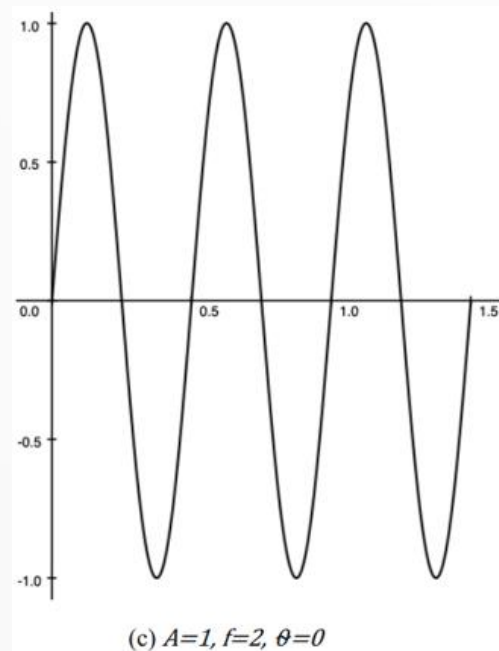
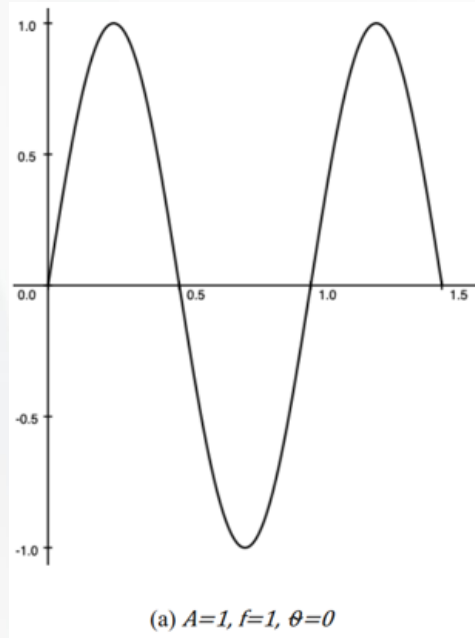
❖ A sine wave, $A \sin(2\pi ft + \theta)$, represents the simplest form of a wave.

- ✓ A = Amplitude
- ✓ f = Frequency
- ✓ θ = Phase
- ✓ t = Current time



Frequency, Wavelength, Amplitude, and Phase

- ❖ Sine waves are cyclic and one complete pattern is a cycle.
- ❖ Frequency, measured in Hertz (Hz), is the number of cycles per second.



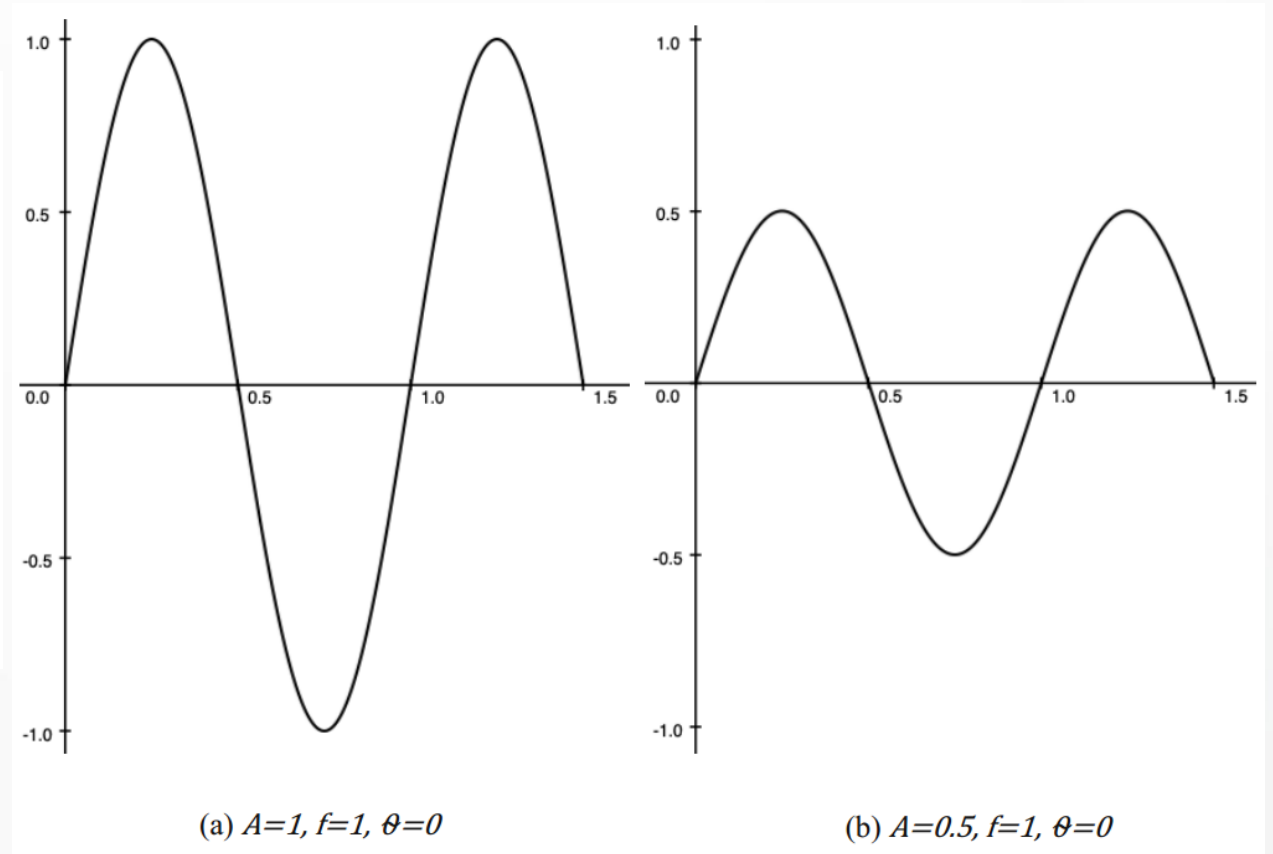
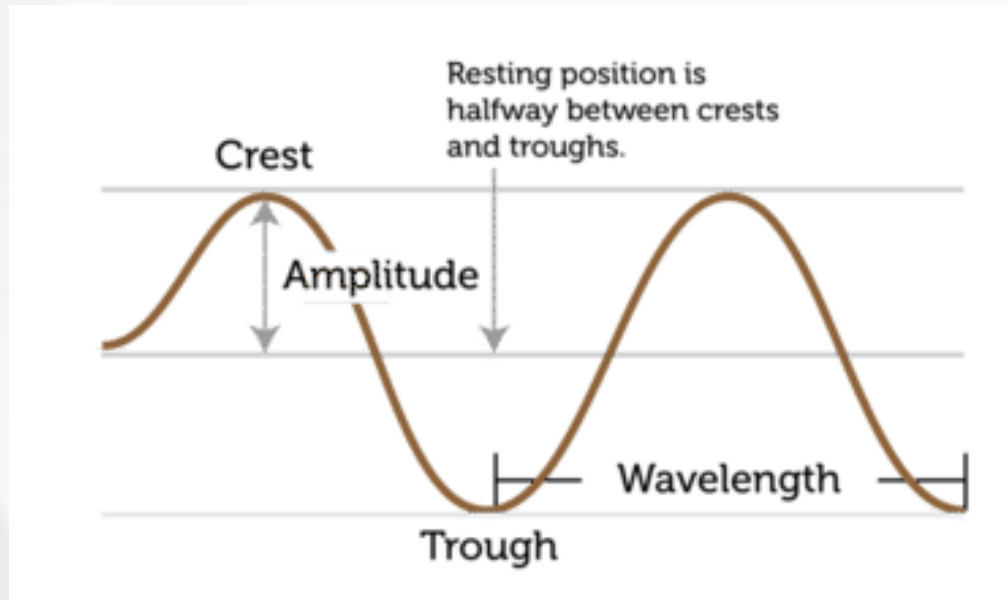
Unit	Abbreviation	Meaning
Hertz	Hz	Cycles per second
Kilohertz	kHz	1000 Hz
Megahertz	MHz	1,000,000 Hz
Gigahertz	GHz	1,000,000,000 Hz

Wave period (T) is just how much time wave takes for one cycle to pass and the units are always in terms of time.

The faster a wave moves, its wave period becomes smaller. **$T = 1/f$**

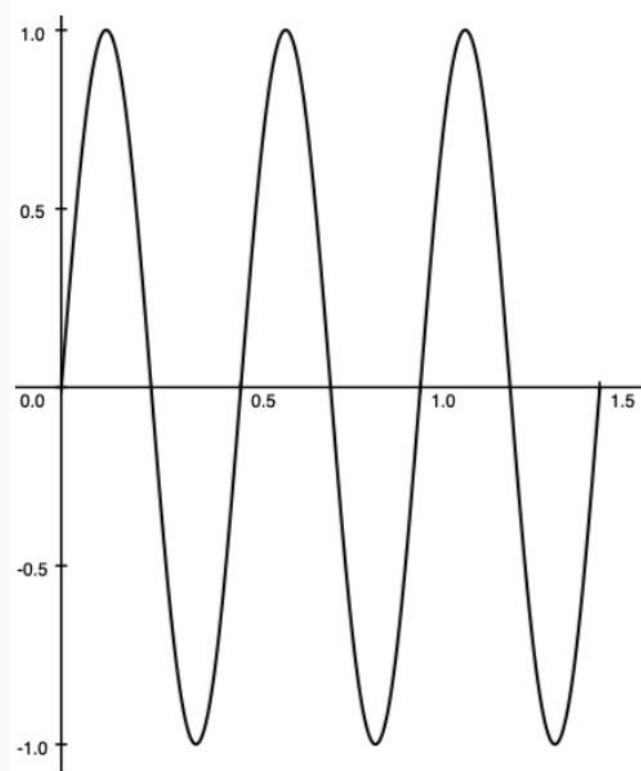
Frequency, Wavelength, Amplitude, and Phase

❖ **Amplitude** is the height of the wave from zero to its maximum value.

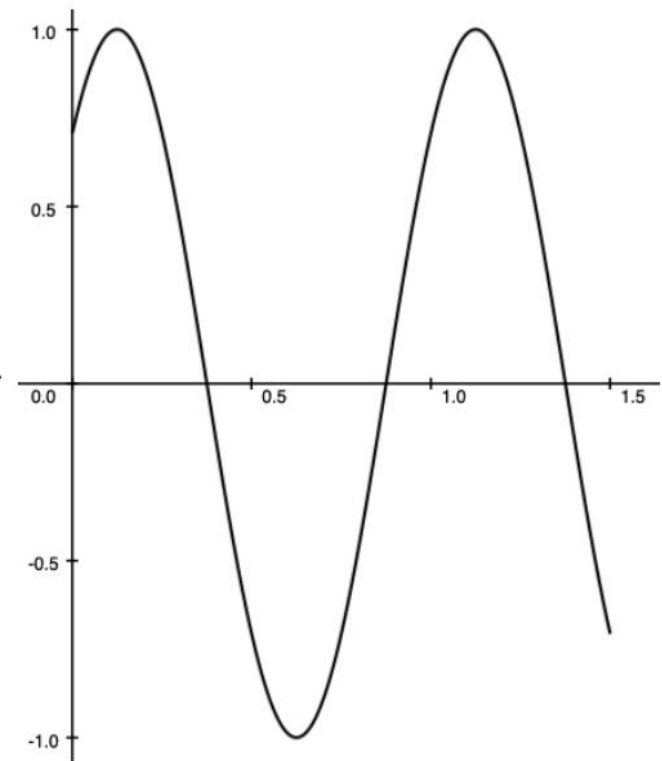


Frequency, Wavelength, Amplitude, and Phase

- ❖ Phase is the shift from a reference point, with zero phase starting at zero amplitude.
- ❖ The maximum phase shift is 360° , equating to a full cycle, returning phase to zero.
- ❖ Phase is usually measured in radians, where $360^\circ = 2\pi$ radians.
 - ✓ *For example, a 45° phase shift equals $\pi/4$ radians.*



(c) $A=1, f=2, \theta=0$



(d) $A=1, f=1, \theta=\pi/4$

Frequency, Wavelength, Amplitude, and Phase

2D Representation of Phase and Amplitude

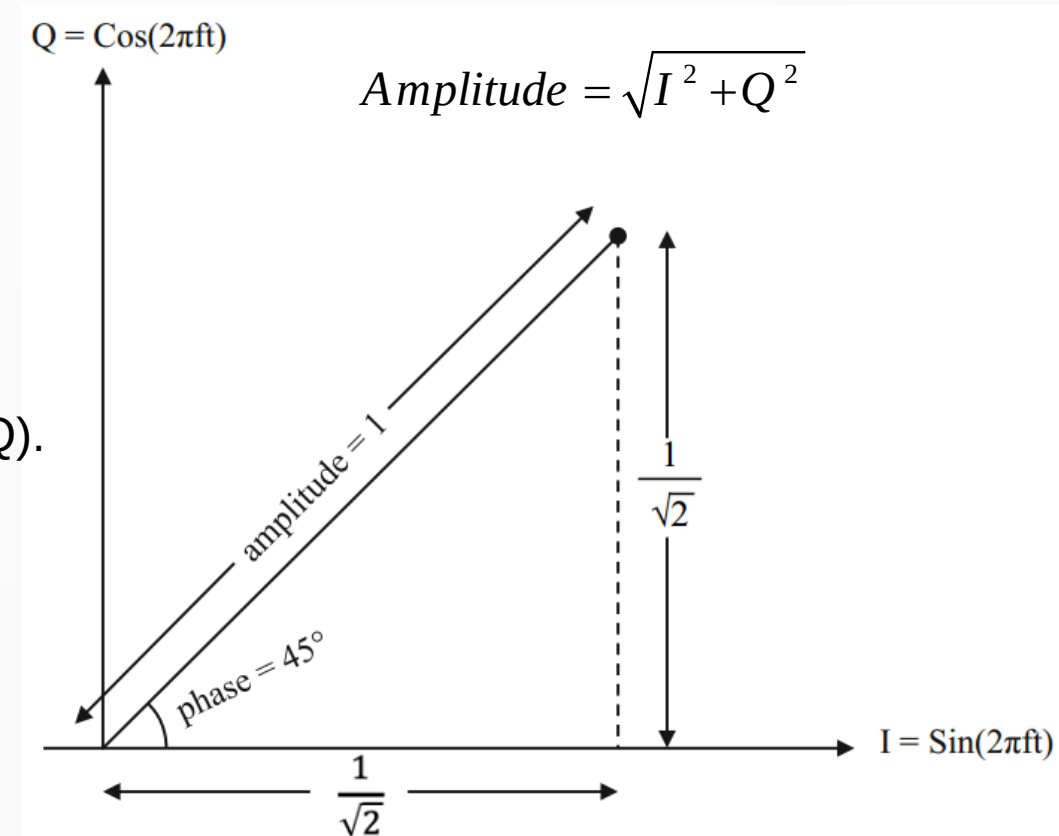
- ❖ For example, a sine wave with a phase of 45° , can be written as the summation of two parts:

$$\begin{aligned}\sin(2\pi ft + \frac{\pi}{4}) &= \sin(2\pi ft) \cos(\frac{\pi}{4}) + \cos(2\pi ft) \sin(\frac{\pi}{4}) \\ &= \frac{1}{\sqrt{2}} \sin(2\pi ft) + \frac{1}{\sqrt{2}} \cos(2\pi ft)\end{aligned}$$

- ❖ The first part is called the In-phase component (I).
- ❖ The second part is called the Quadrature component (Q).
- ❖ In this case:

$$I = \frac{1}{\sqrt{2}} \quad Q = \frac{1}{\sqrt{2}}$$

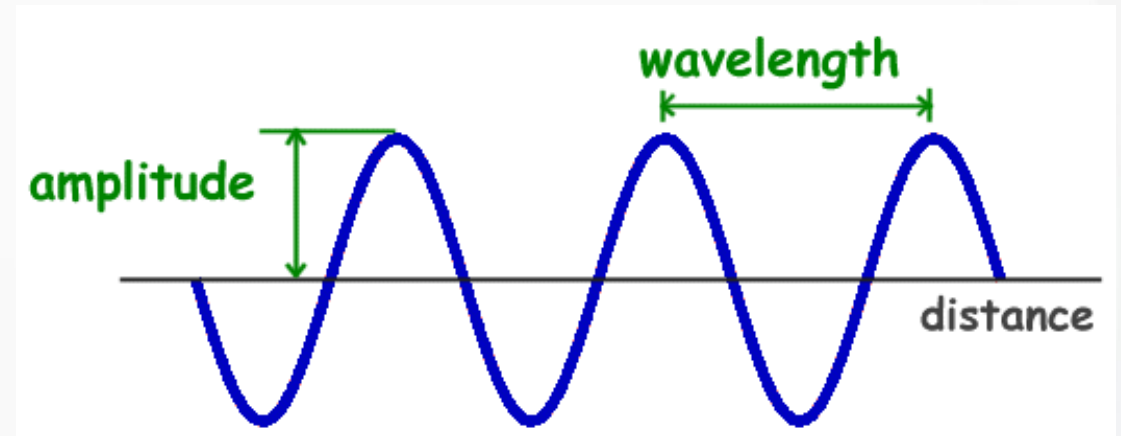
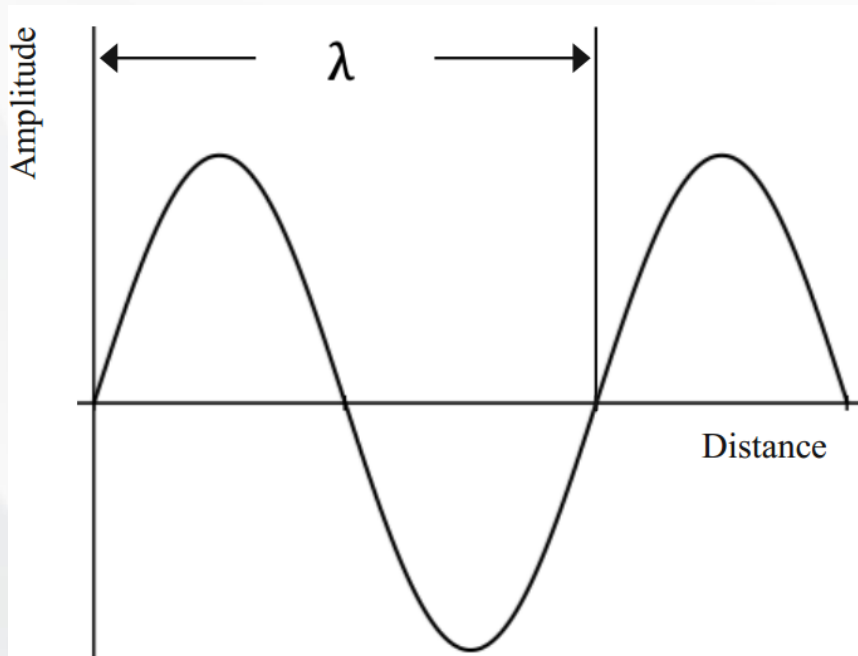
- ❖ I and Q are plotted on a 2D graph.



Frequency, Wavelength, Amplitude, and Phase

Wavelength

- ❖ Waves propagate through space and cover distances over time.
- ❖ The distance occupied by one cycle is called the wavelength (λ).
- ❖ Wavelength is the distance between two points of corresponding phase in consecutive cycles.



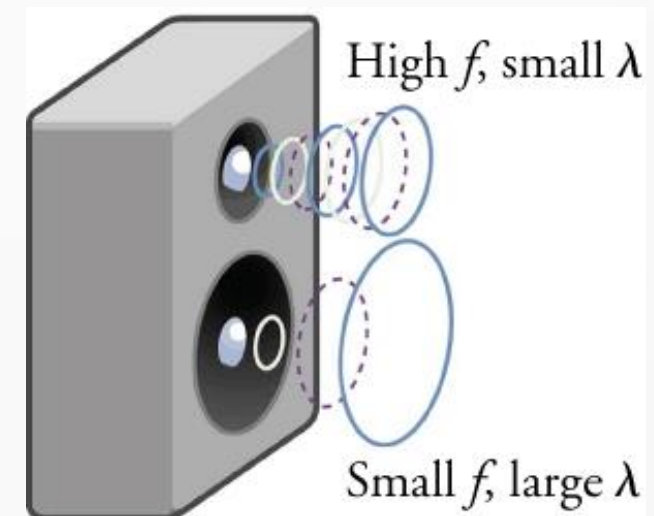
Frequency, Wavelength, Amplitude, and Phase

Wavelength

- In air or space, all electromagnetic waves travel at the speed of light ($300 \text{ m}/\mu\text{s}$) = $3 \times 10^8 \text{ m/s}$.
- The period (T) is the time taken to complete one cycle, with $T = 1/f$.
- Wavelength (λ) is given by the formula

$$\lambda = cT = c/f, \text{ where } c \text{ is the speed of light}$$

- Wavelength is inversely proportional to frequency.



Frequency, Wavelength, Amplitude, and Phase

Wavelength

$$\lambda = c/f$$

- ❖ For acoustic (sound) communications, c represents the speed of sound (343 m/s in dry air at 20° Celsius).
- ❖ Table 1 lists wavelengths for various popular electromagnetic frequencies.

Table 1. Wavelengths of popular electromagnetic frequencies.

Wireless Technology	Frequency	Wavelength
IoT	915 MHz	32.7 cm
Bluetooth/WiFi	2.4 GHz	12.5 cm
Advanced WiFi	700 MHz	42.8 cm
	5 GHz	6 cm
	60 GHz	5 mm
Celluar (3G/4G/5G)	1.7 GHz	17.6 cm
	2.1 GHz	14.2 cm
mmWave and Terahertz for 5G and beyond	28 GHz	1 cm
	73 GHz	4 mm
	140 GHz	2.1 mm
	1 THz	0.3 mm
	10 THz	0.03 mm

Frequency, Wavelength, Amplitude, and Phase

Wavelength

$$\lambda = c/f$$

Example 1

What is the wavelength of a 2.5 GHz electromagnetic signal propagating through air?

Solution

$$\begin{aligned}\text{Wavelength} = \lambda &= \frac{c}{f} \\ &= \frac{300 \text{ m}/\mu\text{s}}{2.5 \times 10^9} \\ &= 120 \times 10^{-3} = 120 \text{ mm} = 12 \text{ cm}\end{aligned}$$

1	=	1e+6
Second		Microsecond

Example 2

What is the frequency of a signal with 5 mm wavelength?

Solution

$$\begin{aligned}\text{Wavelength} = \lambda &= 5 \text{ mm} \\ \text{Frequency} = f &= c/\lambda \\ &= (3 \times 10^8 \text{ m/s})/(5 \times 10^{-3} \text{ m}) \\ &= (300 \times 10^9)/5 = 60 \text{ GHz}\end{aligned}$$

Frequency, Wavelength, Amplitude, and Phase

Wavelength

- Knowing the speed of light, we can calculate the wavelength for a given frequency. Let us take the example of the frequency of IEEE802.11b wireless networking, which is:

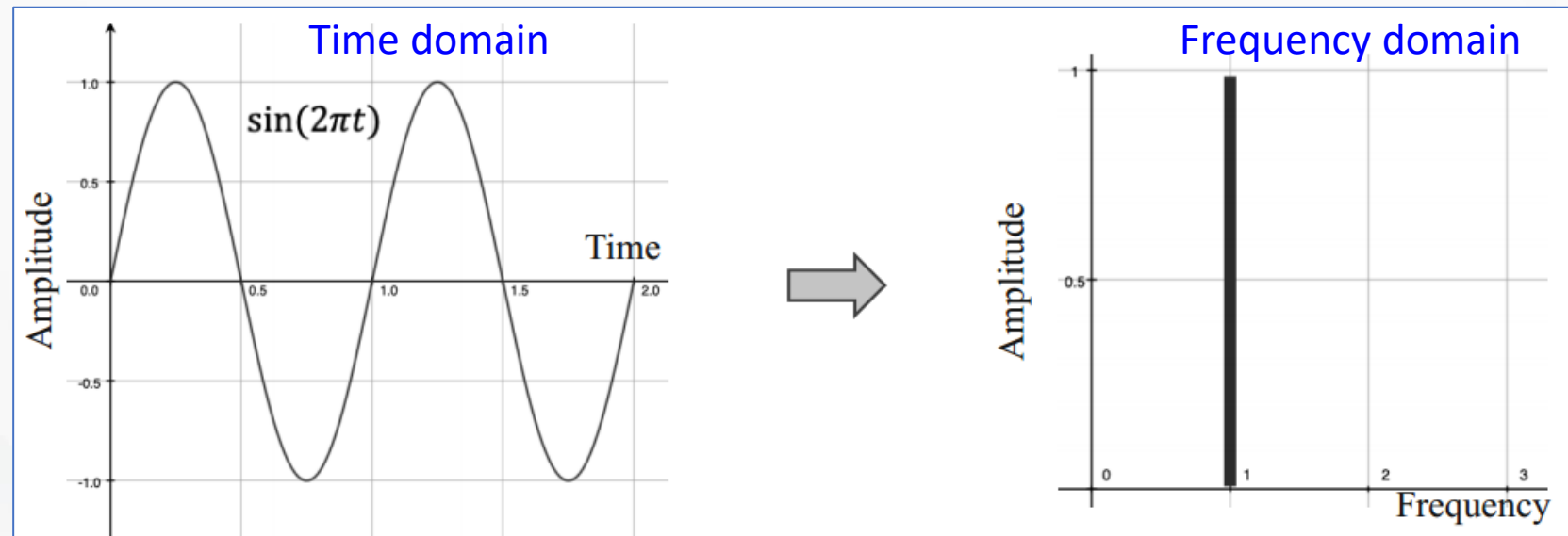
$$f = 2.4 \text{ GHz} = 2,400,000,000 \text{ cycles / second}$$
$$\text{wavelength } (\lambda) = c / f = 3 \times 10^8 / 2.4 \times 10^9 = 1.25 \times 10^{-1} \text{ m} = 12.5 \text{ cm}$$

- Wireless standards, for example, 802.11b, 802.11g, 802.11n and 802.16 can all work at 2.4 GHz

In practice:
Antennas work better
with size = $1, \frac{1}{2}, \frac{1}{4}$ of wavelength
(try to measure antenna size of
your IEEE 802.11 device)

Time and Frequency Domains

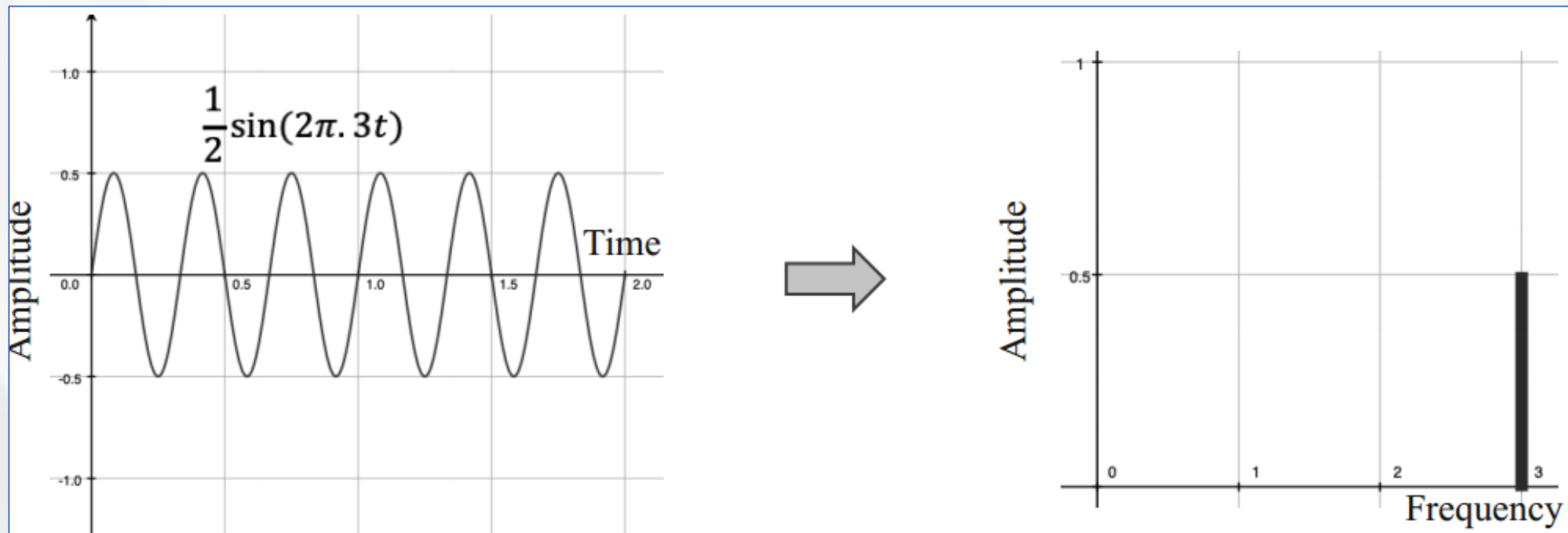
- ❖ Waves can be represented in both **time** and **frequency** domains.
- ❖ Time domain representation can be converted to frequency domain and vice versa.
- ❖ The below Figure illustrates this conversion using three different sine waves.
- ❖ Top sine wave: frequency 1 (f) and amplitude 1 (A) → In frequency domain: pulse at $f = 1$ with height $A = 1$.



Time and Frequency Domains

❖ Second sine wave: frequency $3f$ and amplitude $1/2A \rightarrow$

In frequency domain: pulse at $f = 3$ with height $A = 0.5$.



Time and Frequency Domains

- ❖ **Fourier transform** : The transformation from time to frequency domain.
- ❖ **inverse Fourier transform** : The reverse transformation.
- ❖ Fast algorithms: include Fast Fourier Transform (**FFT**) and Inverse FFT (**IFFT**).
- ❖ **Packages** like MATLAB and programming languages like Python.

```
import numpy as np
import matplotlib.pyplot as plt

# Generate a sample signal: a sum of two sine
waves
Fs = 500 # Sampling frequency
T = 1 # Duration in seconds
t = np.linspace(0, T, int(Fs * T),
endpoint=False) # Time vector

# Frequencies of the sine waves
f1 = 5 # 5 Hz
f2 = 50 # 50 Hz
```

Time and Frequency Domains

- ❖ **Fourier transform** : The transformation from time to frequency domain.
- ❖ **inverse Fourier transform** : The reverse transformation.
- ❖ Fast algorithms: include Fast Fourier Transform (**FFT**) and Inverse FFT (**IFFT**).
- ❖ **Packages** like MATLAB and programming languages like Python.

```
# Sample signal
```

```
signal = 0.5 * np.sin(2 * np.pi * f1 * t) + 0.2 *  
np.sin(2 * np.pi * f2 * t)
```

```
# Compute the FFT
```

```
fft_result = np.fft.fft(signal)  
fft_freqs = np.fft.fftfreq(len(fft_result), 1 /  
Fs)
```

```
# Only keep the positive frequencies
```

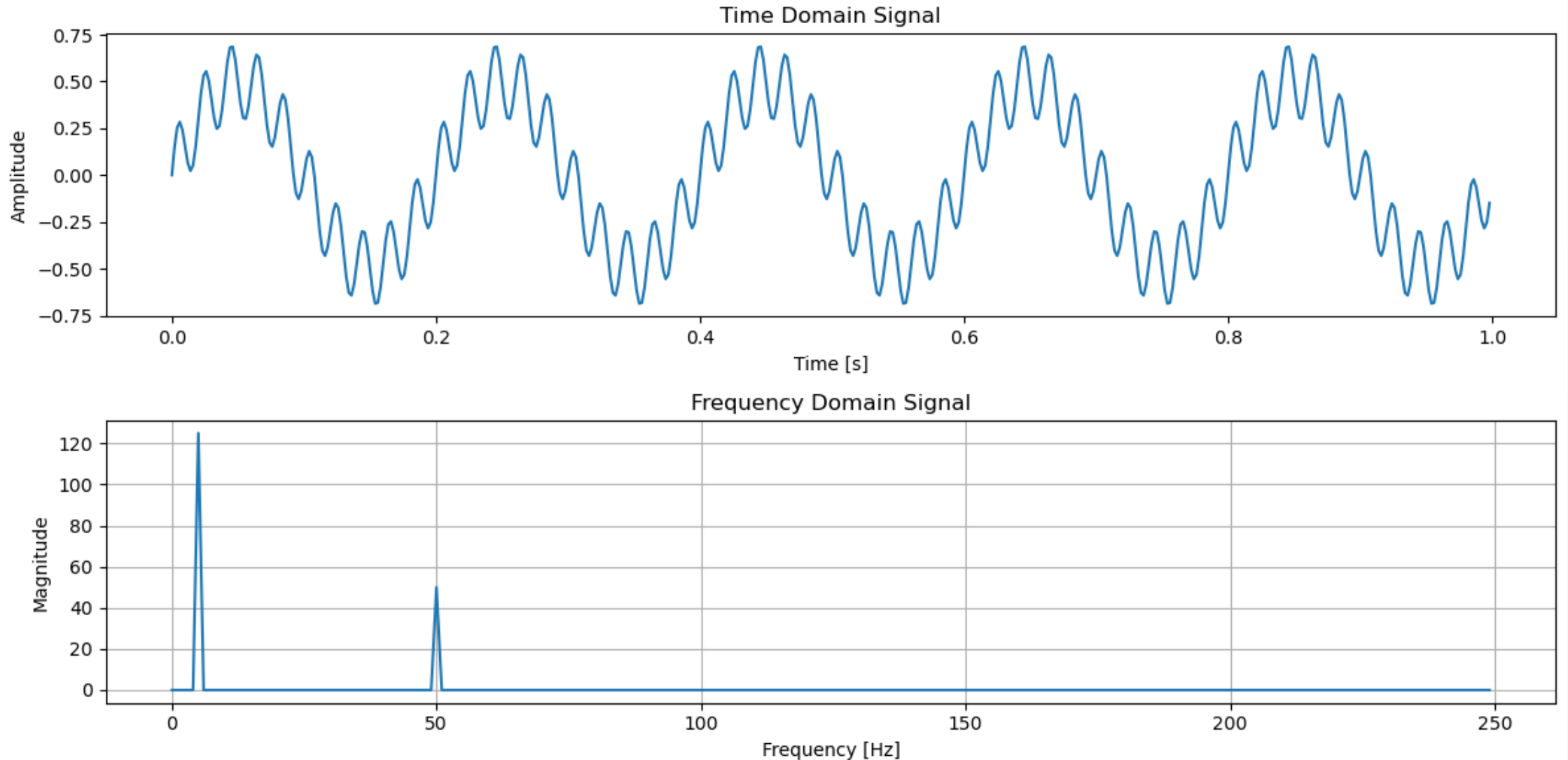
```
positive_freqs = fft_freqs[:len(fft_freqs) // 2]  
positive_magnitudes =  
np.abs(fft_result[:len(fft_result) // 2])
```

Time and Frequency Domains

```
# Creates a new figure with a size of 12 inches in width and 6 inches in height.
plt.figure(figsize=(12, 6))

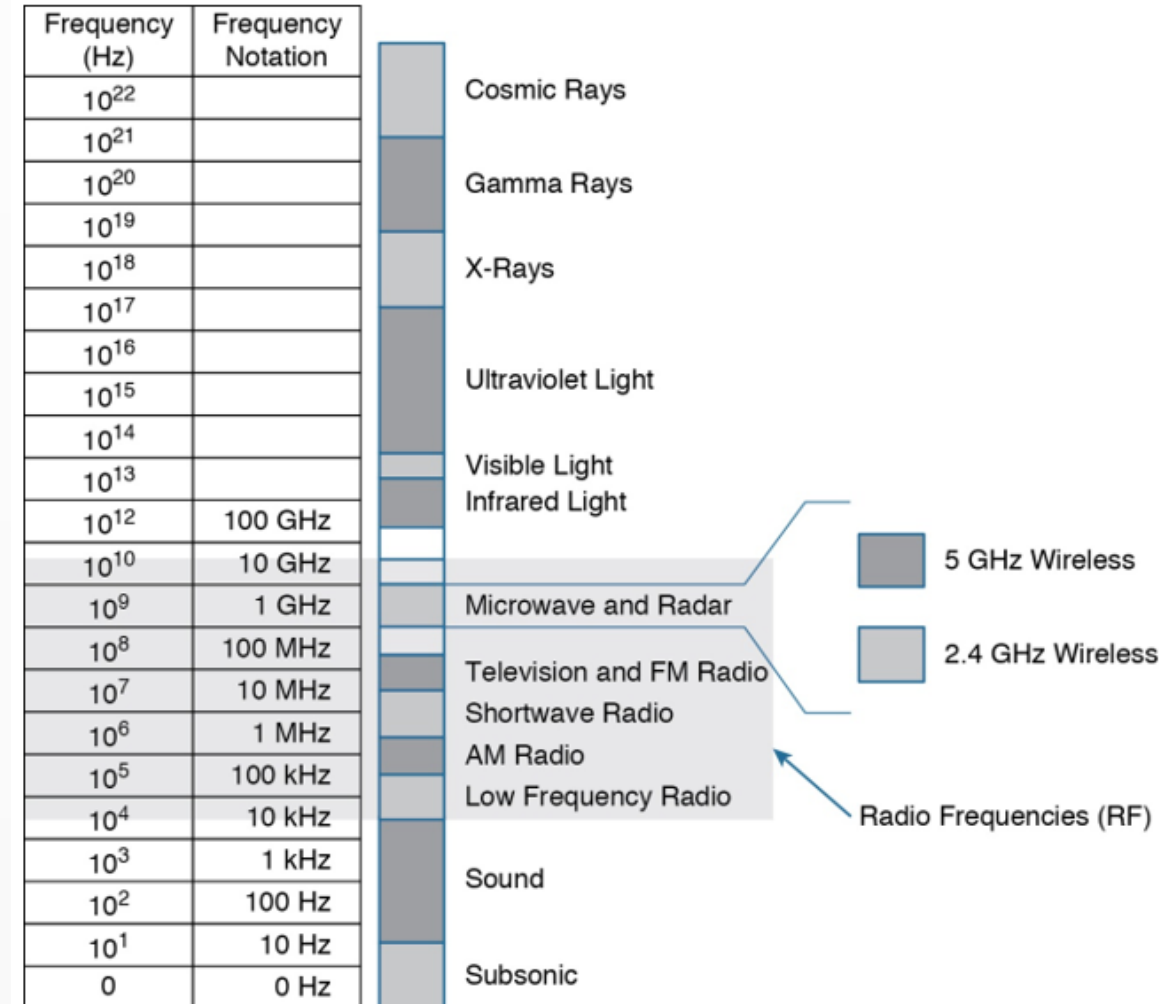
#creates the first subplot in a 2-row by 1-column grid (i.e., two subplots stacked
vertically). The 1 at the end specifies that this is the first subplot.
plt.subplot(2, 1, 1)
plt.plot(t, signal)
plt.title('Time Domain Signal')
plt.xlabel('Time [s]')
plt.ylabel('Amplitude')
# Plot the signal in frequency domain
plt.subplot(2, 1, 2)
plt.plot(positive_freqs, positive_magnitudes)
plt.title('Frequency Domain Signal')
plt.xlabel('Frequency [Hz]')
plt.ylabel('Magnitude')
plt.grid()
plt.tight_layout()
plt.show()
```

Time and Frequency Domains



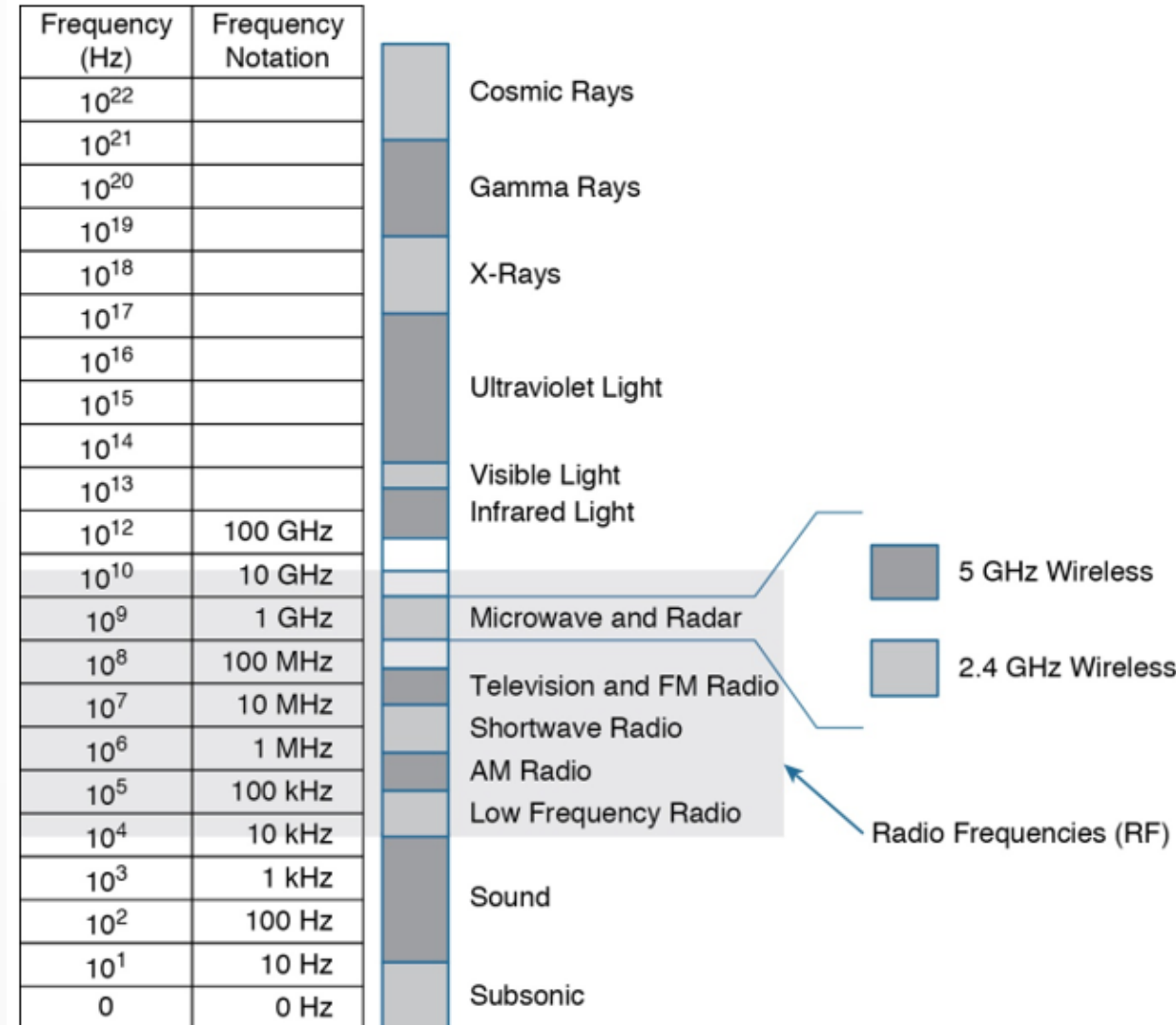
Electromagnetic Spectrum

- ❖ Wireless communications use **electromagnetic** waves that can propagate through **air** or a **vacuum**.
- ❖ Many devices generate or use **electromagnetic** waves: TVs, power supplies, remote controls, microwave ovens, wireless routers, etc.
- ❖ Electromagnetic waves span a wide frequency range, from **10 Hz to 300 THz**.
- ❖ The spectrum comprises all usable frequency ranges but its **limited** natural resource.
- ❖ Spectrum use is regulated by authorities like the **FCC** in the US and **ACMA** in Australia.



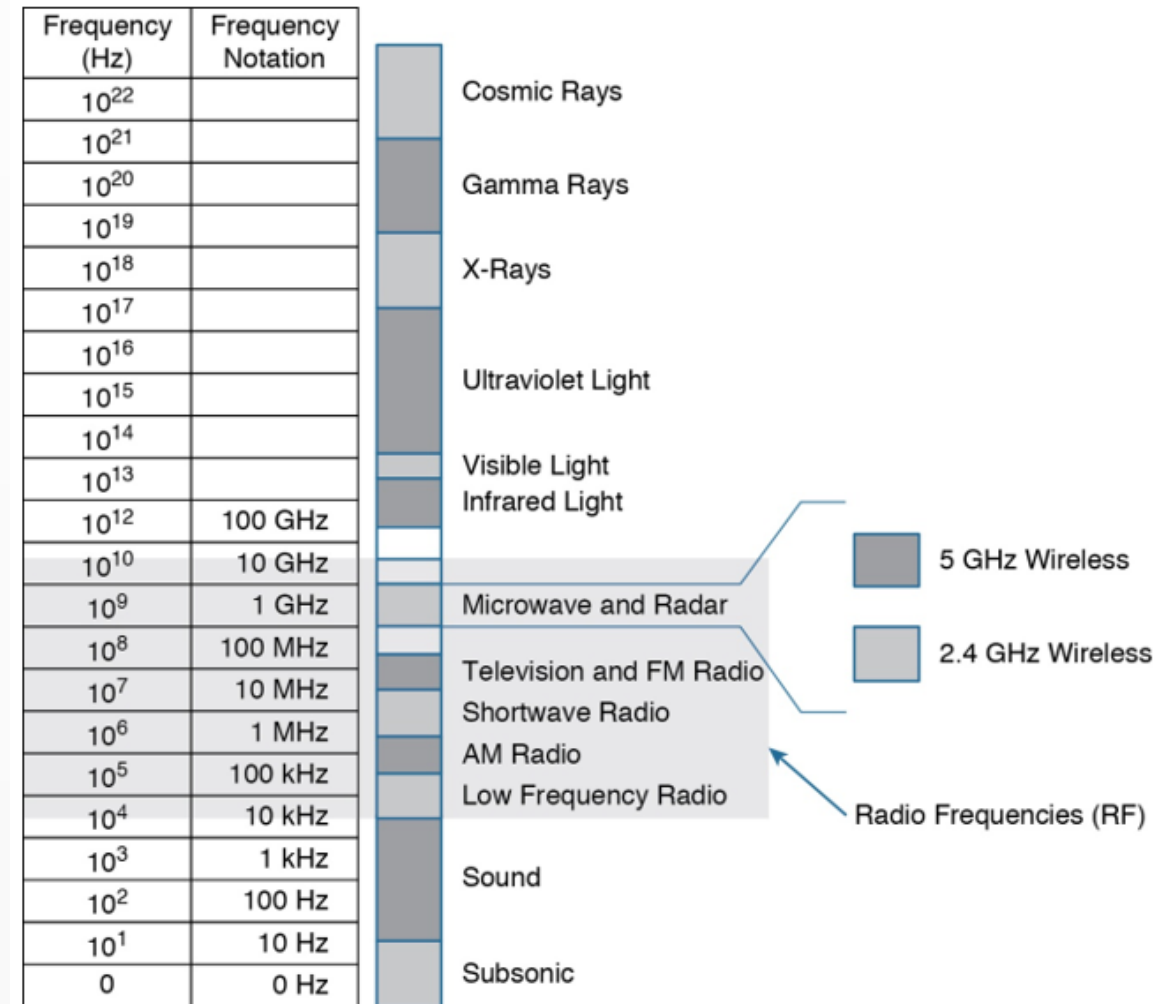
Electromagnetic Spectrum

- ❖ Large portions of the spectrum are reserved for **government** use (e.g., radar, military communications).
- ❖ Remaining spectrum is licensed to network operators for exclusive use
 - ✓ (e.g., TV channels, radio stations).
- ❖ Spectrum-allocation authorities follow principles to **maximize utilization**, promote **new technologies**, ensure market competition, fairness, and satisfy national interests



Electromagnetic Spectrum

- ❖ Historically, wireless communications have used the spectrum between 100 kHz to 6 GHz.
- ❖ The industry is exploring spectrum beyond 6 GHz to meet mobile data demand (e.g., 60 GHz for some Wi-Fi standards, 5G targeting beyond 60 GHz).

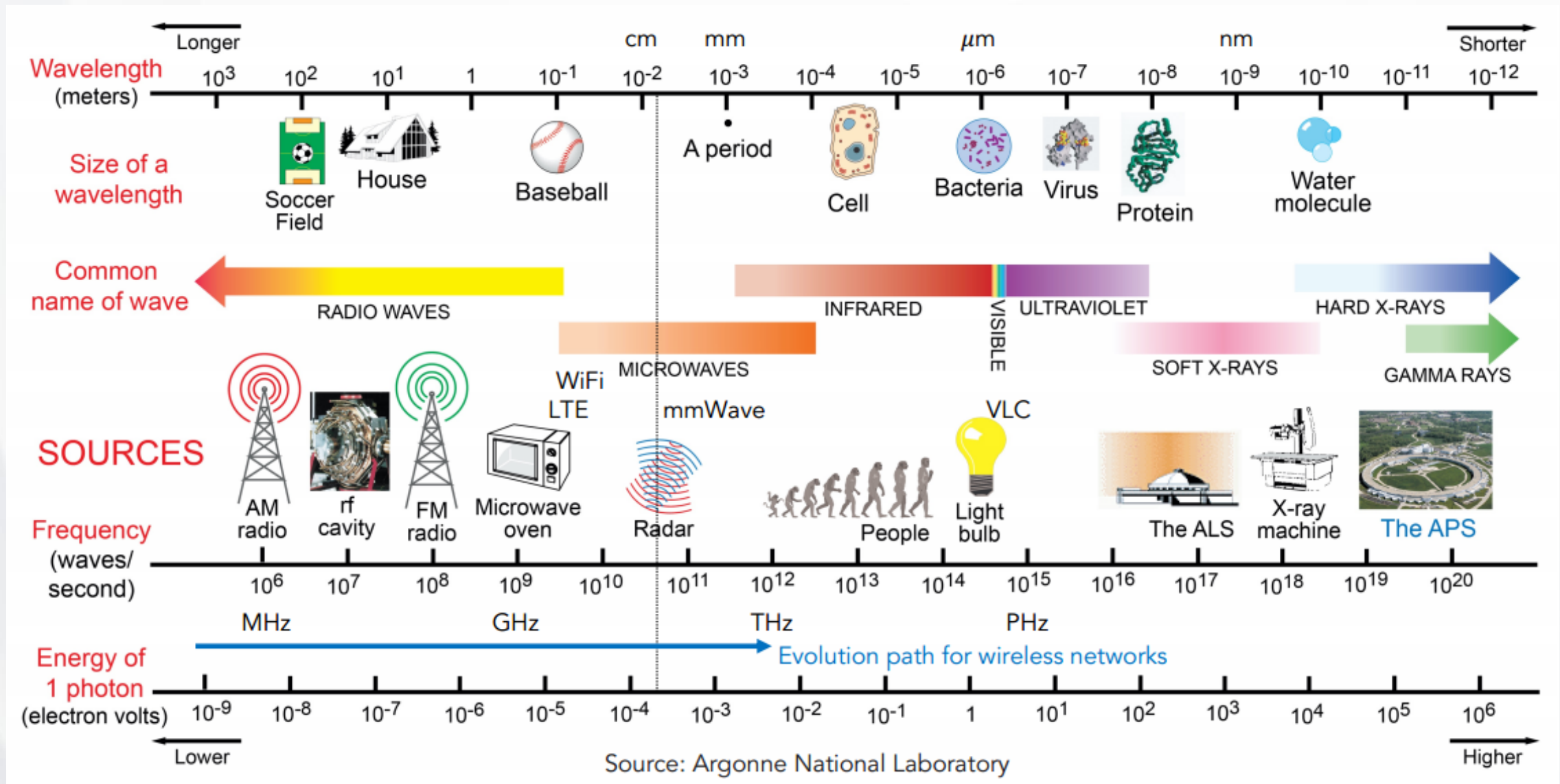


Electromagnetic Spectrum

- ❖ Some spectrum is **unlicensed**, allowing free use (e.g., 2.4 GHz band for Wi-Fi).
- ❖ **Unlicensed** spectrum usage is subject to rules, like power limitations (e.g., Wi-Fi transmit power is limited to about 100 mW).

License-exempt Spectrum	Example Use
433 MHz	Keyless Entry
900 MHz	Amateur Radio, IoT (e.g., LoRaWAN)
2.4 GHz	WiFi, Microwave Oven
5.2/5.3/5.8 GHz	WiFi, Cordless Phone

Electromagnetic Spectrum



THANKS

